QA-CANDIDATE

QA candidate: build it, run it, test every flow

Field	Value
Revision	1
Created	2026-10-22
Last modified	2026-10-22
Status	active (the candidate is rebuilt by script as streams advance; this page describes the mechanism, the generated manifest is the record of what was in a given build)
Audience	the operator and the manual-QA testers
Scripts	`scripts/qa-candidate.sh` (build the candidate branch), `scripts/qa-candidate-up.sh` (run the isolated instance)

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1. What the candidate is, and what it is not

The QA candidate is an **integration** of everything that has been built so far, including branches that are still in review or fix rounds. It exists so that you can use every flow and report what is wrong.

- It is **not a release**, **not main**, and **not reviewed as a whole**. Expect defects. Finding them is the purpose.
- It lives on its own branch, `qa-candidate`, in its own git worktree. Nothing is ever merged from it into `main` or `int-0.0.3`, and nothing is pushed anywhere by the scripts (`scripts/qa-candidate.sh` contains no push and refuses any push-shaped argument).
- The branch is rebuilt **by script** (`scripts/qa-candidate.sh`) from the committed tips of the stream branches. The record of what went in, at which commit, and how any conflict was resolved is `evidence/qa-candidate/MANIFEST.tsv` inside the candidate branch.
- The instance you test is a **plain server process on its own port** with a **copy** of the live data. It shares nothing writable with the live service.

2. Start, stop, reset

All commands run from the umbrella checkout (`~/Projects/vasic`).

Task	Command	Notes
First start (everything)	`bash scripts/qa-candidate-up.sh up`	Builds the server and the Angular bundle from the candidate worktree, copies the data, applies the research-library re-ingest to the copy, provisions the seven accounts, starts the server and waits until it is healthy. The first run takes several minutes (the Angular build waits for the host's load governor).

Task	Command	Notes
Is it running?	`bash scripts/qa-candidate-up.sh status`	Exit 0 up and answering, 1 down. Prints the URL.
Health check	`bash scripts/qa-candidate-up.sh health`	`/api/health`, then a sign-in as each administrator and as `student1`. Prints status codes only, never a password.
Stop	`bash scripts/qa-candidate-up.sh down`	Signals only the process this script started (verified by its executable).
Reset the QA data	`bash scripts/qa-candidate-up.sh reset`	Stops the server, wipes **only the QA copy** (`data/`), copies the live data again, re-applies the research re-ingest, re-provisions the accounts. Students go back to the initial password. The binaries and the credentials file are kept.
Rebuild binaries and bundle after the candidate branch moved	`bash scripts/qa-candidate-up.sh down && bash scripts/qa-candidate-up.sh build && bash scripts/qa-candidate-up.sh up`	`build` always recompiles the Go binaries; it recompiles the bundle only when no bundle exists in the worktree (delete `platform/frontend/dist` in the worktree to force it).
LAN access	`QA_CANDIDATE_BIND=0.0.0.0 bash scripts/qa-candidate-up.sh up`	Off by default. The instance serves a recording of an identifiable third party; binding

Task	Command	Notes
		beyond loopback exposes it to the network. The script says so before binding.

If the instance goes down (host restart, session ended, someone ran `down`): `bash scripts/qa-candidate-up.sh up` is idempotent and starts it again with the same data. To restart automatically run `setsid nohup bash scripts/qa-candidate-up.sh watch >/dev/null 2>&1 &` (checks `/api/health` every 30 s, runs `up` when it fails; stop it by killing that one pid).

Individual stages can be run on their own: `build`, `data`, `research`, `accounts`. Environment knobs (`QA_CANDIDATE_QA_ROOT`, `QA_CANDIDATE_PORT`, `QA_CANDIDATE_BIND`, `QA_CANDIDATE_CREDS`, `QA_CANDIDATE_OLLAMA`, `QA_CANDIDATE_SKIP_WEB`, `QA_CANDIDATE_NG_WRAPPER`) are documented in the header of the script (`bash scripts/qa-candidate-up.sh --help`).

3. Where it runs, and what it cannot touch

Item	Value
URL	`http://127.0.0.1:<port>`; the port is chosen at start (first free in 18300..18399, **never 8087**) and printed by `up` / `status`. The last start used **18300**.
Process	one `workshop-server` started by `qa-candidate-up.sh`, no container, no compose
Instance root	`~/.cache/workshop-qa-candidate/` (`bin/` binaries, `data/` the QA data COPY, `state/` pid, port, log, staged web bundle)
Server log	`~/.cache/workshop-qa-candidate/state/server.log`
Answering	**OFF** (`-answer-provider none`); Ask answers that it is not configured; browsing and search work
Hosted providers	OFF; no `ApiKey_*` variable and no API-key file reach the process; nothing billable can be called
Semantic search	uses the local Ollama embedding endpoint (read-only, non-billable) when it is up; set `QA_CANDIDATE_OLLAMA=` (empty) for lexical-only

Item	Value
Live service	never touched: the live registry, the live checkout and `:8087` are only READ to make the copy. `up` refuses port 8087 and any QA root inside the live checkout

Data copy. `data/curriculum` (registry `passages.jsonl`, the three textbooks' catalogue and sidecars, learning catalogue, questions), `data/docs` (including `docs/research`), `data/chapters` (recordings and transcripts, without the archive part files) and `data/prebuilt`. **Research library:** the five research documents that had no registry rows (`clients/jordan` and the four `education-platform/*` documents) were staged and applied to the COPY with `scripts/ingest-research-docs.sh` (120 passages added, 0 existing rows changed, 0 model calls measured), so they are searchable. **Not included:** diagrams (the textbooks therefore show no figures).

Semantic index note. The shipped prebuilt vector bundle was built over a different corpus than the QA copy (the copy now carries the 120 research passages), so the server logs "prebuilt vectors: nothing imported" and builds its own index; a search reports `legs: lexical ok, semantic ok`.

4. Accounts and credentials

Seven accounts exist in the QA auth store (spec 023): administrators (professors) **`milosvasic`** and **`rami`**; students **`student1`** .. **`student5`**.

- The passwords are **random 20-character values generated for this instance**. They are in **`~/.cache/claude-tmp/qa/credentials.env`** (mode 0600). The file has three entries: `WORKSHOP_ADMIN_PASSWORD_MILOSVASIC`, `WORKSHOP_ADMIN_PASSWORD_RAMI`, `WORKSHOP_STUDENT_INITIAL_PASSWORD` (one initial password shared by the five students). Open the file yourself; the scripts never print a value.
- The operator's own weak initial student password is not known to these scripts and is not used.
- **Students start flagged "must change password":** after the first sign-in the server allows only `/account` (`403 password_change_required` everywhere else). That is a flow to test (spec 023); after the change the student signs in with the new password. `reset` returns students to the initial password.

- Sign-in is rate limited (5 attempts per minute per account); a "too many login attempts" answer is the limiter working, wait a minute.

5. What is in the candidate

Built from `int-0.0.3` by `git merge --no-ff` of each stream's committed tip, in the order of `scripts/qa-candidate.branches.tsv`. The table is the generated `evidence/qa-candidate/MANIFEST.tsv` of the build described here (candidate HEAD `bb9bfd7da0d3`).

Branch	Merged tip	Status	Notes
`base`	`int-0.0.3`	base	35f2ea38e768
`main`	`d0da0a7a476e`	conflict-resolved	merge commit b1c9459f
`fix-apikeys-argv`	`11f2bb5ea02a`	merged	merge commit 871dacc0
`wire-completion`	`765f40e2e72b`	conflict-resolved	merge commit f0b79314
`n1-013-adaptive`	`c7f95645bce6`	merged	merge commit d76e5840
`n8-013-ui-gate`	`9468906e99a7`	conflict-resolved	merge commit e96ff431
`n3-020-024-complete`	`bf2232eeec50`	merged	merge commit a8b112cf
`n5-011-014-015-complete`	`18d9eda64076`	merged	merge commit 1df0531c
`n4-016-complete`	`545777bbc72b`	conflict-resolved	merge commit c45e4c4f
`n2-012-ingest-complete`	`760066c4d00e`	merged	merge commit 9fd18c7f
`s9-014-modal-polish`	`2fef67bfa034`	conflict-resolved	merge commit 198cb4c5
`s11-018-library`	`5e826bb17a9d`	conflict-resolved	merge commit 6c5bcc33
`s12-019-promotion`	`24c22007954a`	conflict-resolved	merge commit 88881c91
`s15-025-ux`	`b344ae7bf7a6`	conflict-resolved	merge commit 27891737
`s16-021-medium`	`905bf091b7af`	conflict-resolved	merge commit 146cd2a4
`s17-022-material`	`44dbdb8b0c00`		merge commit 10f31161

Branch	Merged tip	Status	Notes
		conflict-resolved	
`s18-023-roles`	`5adb94f51c80`	conflict-resolved	merge commit ea276979
`cat-authors`	`e0d274353471`	merged	merge commit 5618228c
`h1-providers-hardening`	`126e4cc0dd1c`	conflict-resolved	merge commit fd4b9936
`inv-nonascii`	`51f40ff14410`	merged	merge commit 32cdeaef
`s2-016us1-013us1`	`7496f52b8745`	merged	merge commit 3aef7055
`n6-001-002-complete`	`a5847e720e21`	merged	merge commit 02304a18
`n7-integration`	`002bd428289c`	conflict-resolved	merge commit f4a34956
`qa-fix`	`7dff042c`	qa-fix	qa-fix: achievement-badge-producer.spec builds a ProgressState with the ability_sco field n1 made required (merge-induced sp compile error)
`qa-fix`	`c79ff4b0`	qa-fix	qa-fix: gofmt ratchet - n7 formatted the 13 baselined files (baseline emptied) and 3 fil from other streams were unformatted (go -w); gate verify
`qa-fix`	`56c8d516`	qa-fix	qa-fix: TestLazyHostedBuildsOnlyOnFirstGenerat drives Generate with the per-request acknowledgement n3's remote_spend now requires (merge-induced t
`qa-fix`	`7020948a`	qa-fix	qa-fix: wire_materialgen lazy-hosted test a installs the spend hook h1's billable probe requires (the earlier ack-only change was enough); real
`qa-fix`	`a92f80d5`	qa-fix	qa-fix: remove the name-only ApiKey_OpenAI/ApiKey_Anthropic compo environment entries that my n8 conflict union re-introduced (apikeys design: keys
`qa-fix`	`b1c4b744`	qa-fix	qa-fix: gofmt main.go (union of wire-completion + n7 edits)
`qa-fix`	`d0b0cc1d`	qa-fix	qa-fix: docs/faq.md entries renumbered 19..33 sequentially (the unions left several

Branch	Merged tip	Status	Notes
			duplicate numbers); one cross-link update exports not regenerate

Rows with status `conflict-resolved` needed a human union of the streams' registrations, routes, flags and docs; the resolution notes are in the merge commit message of each. Rows `qa-fix` are small commits made on the candidate for merge-induced breakage only. Product defects are NOT fixed on the candidate; they are in section 7.

What was verified on this build (this session)

Check	Result
`go build ./...`, `go vet ./...` (platform/backend)	clean
`go test -count=1 ./...` (platform/backend, throwaway seed passwords)	every package ok, after the three `qa-fix` test repairs listed in the manifest
`tsc --noEmit` on the app and spec projects	clean
`ng build`	succeeded, 0 errors (bundle warnings only)
Karma (`ng test --watch=false --browsers=ChromeHeadless`)	queued behind the host load governor and NOT completed at the time of writing: UNVERIFIED
Gates (22 run directly, see KNOWN-DEFECTS part 1)	16 PASS; `verify-suppressed-residue` and `verify-redaction-propagation` FAIL (CC-01, CC-02); 4 UNDETERMINED for want of untracked data (CC-12)
Instance	started on 127.0.0.1:18300; `/api/health` 200; sign-in as `milosvasic`, `rami`, `student1` 200; anonymous `/api/areas` 401; a student on `/api/admin/users` 403 and on `/api/areas` 403 `password_change_required` until the password is changed; library lists 9 documents (research incl. `clients/jordan` and `education-platform`); textbooks 3 books / 1675 sections / 0

Check	Result
	diagrams; search legs lexical ok, semantic ok
Browser-driven UI testing	NOT done by the builder: nothing in section 8 was executed through a browser

6. Rebuilding the candidate after branches advance

```
bash scripts/qa-candidate.sh            # merges only the tips
        that are not yet contained; idempotent; rc 0 / 1 / 2
bash scripts/qa-candidate.sh --status   # read-only: what a run
        would do
bash scripts/qa-candidate.sh --prove-failure   # paired proof of
        the script's own guarantees
```

Exit **0** every listed branch is merged and the candidate builds; **1** a branch is missing, conflicted, or was rejected for breaking the Go build (it is aborted and recorded in the manifest; the candidate stays buildable); **2** could not determine (git missing, dirty candidate worktree, forbidden argument). A conflict is resolved once by hand in the candidate worktree (union; commit subject ending `[conflict-resolved]`); `git rerere` is enabled there so the same resolution replays. A branch that cannot be merged without breaking the build is parked by adding a reason in the third column of its row in `scripts/qa-candidate.branches.tsv`. After the branch moved, restart the instance with `down`, `build`, `up` (section 2).

7. Known defects and limits

The full list is `evidence/qa-candidate/KNOWN-DEFECTS.md` in the candidate branch (also published next to this page as `docs/QA-CANDIDATE-KNOWN-DEFECTS.md`). It combines (a) the open findings of the independent reviews, each with its status in this build, and (b) the problems found while constructing the candidate. Rows in the checklist below marked `KNOWN-DEFECT` point into it or into the review that raised the finding. The candidate is expected to have these; report anything else.

8. Manual QA checklist, spec by spec

How to read it. One table per spec. *Role* is who performs the step: anonymous, student, professor-admin (`milosvasic` / `rami`), or operator (a command on the host). *Status*: **TESTED-BY-AUTOMATION** names the file that exercises the flow (many e2e specs run against mocked responses or need a special server: those are marked `partial`); **NOT-AUTOMATED** means a human must look; **KNOWN-DEFECT** points to the finding. A row marked "verify" is a place where the spec was read but the screen was not: tell us what you see. These tables were drafted from the specs and the candidate source tree; **no row has been executed by a person yet**, and no row claims a pass. Evidence paths are relative to the candidate worktree (`~/Projects/vasic-wt/qa-candidate`) unless they start with `/`.

Flags. All eight spec-025 UX features default OFF and are enabled per browser (localStorage key `workshop-ux-flags`); on-demand material generation (022) is not mounted in this build.

After the research re-ingest on the QA copy the five research documents ARE searchable (a row that says "Not searchable yet" describes the un-ingested live registry, not this instance).

Part A

Candidate tree: `~/Projects/vasic-wt/qa-candidate` (all paths below are relative to it). QA instance: answering OFF, hosted providers OFF. Accounts: administrators (professors) milosvasic and rami, students student1..student5. No credentials appear here.

Conventions

- `<slug>` for an area comes from the area grid on `/` (Knowledge areas). `<pid>` is a 26-character passage id copied from a search hit or transcript row. `01` is the first chapter id.
- "e2e" specs live in `platform/frontend/e2e/`. Several run against a live stack, so citing one means the flow has an automated check, not that it was run against this QA instance.
- Evidence for KNOWN-DEFECT rows is `platform/gates/defects-registry.tsv` (ids quoted). Those entries are dated; the tester should confirm the defect still reproduces on this build. No file in `the review reports (by file name)` mentions specs 001 to 004 content (checked by grep); those reviews cover branches 011 to 026 and the constitution governor.

- API-only checks need a signed-in session token supplied by the operator. Do not paste it anywhere.

Spec 001 — Workshop curriculum platform

Delivers a browsable curriculum of recorded chapters (transcripts with timestamps, recording playback, materials), meaning-based search with suggestions, and cited question answering. Also covers the add-a-chapter procedure, redaction, accessibility (WCAG 2.1 AA) and the evidence discipline. With answering OFF on this instance, FR-021 to FR-024 and FR-053 cannot be exercised end to end; only the "answering is off, browsing and search still work" behaviour (FR-025) can.

ID	Role	Flow / click path (URL)	Expected result	FR / SC
QA-001-01	anonymous	Open `/` and `/chapters/01` signed out. Also `curl -i <host>/api/chapters` with no token.	Browser lands on `/login?returnUrl=...`. API returns 401, no chapter, transcript or search content is served.	FR-051
QA-001-02	student	Sign in as student1, open `/`.	Curriculum front door: "An index of a recording." heading, corpus figures ("What is indexed"), one expandable rail per chapter (first expanded), Knowledge areas grid, Chapters grid.	FR-008, FR-011
QA-001-03	student	Open `/chapters`.	Every chapter the API reports is listed with	FR-008

ID	Role	Flow / click path (URL)	Expected result	FR / SC
			title, summary and ordinal; reduced payloads are labelled "reduced", not padded with zeros.	
QA-001-04	student	Open `/chapters/01`.	Chapter detail: title, summary, recording player, supporting materials labelled as supporting, link to transcript. Archive and checksum manifest are NOT listed as materials.	FR-009, FR-011, FR-006
QA-001-05	student	On `/chapters/01` press play, pause, seek with arrow keys, then "jump to this timestamp", mouse and keyboard only.	Recording plays and seeks (Range requests honoured); every control is Tab-reachable, Enter/Space-activatable and has an accessible name.	FR-009, FR-059
QA-001-06	student	Open `/chapters/01/transcript`; scroll the whole page.	Full transcript from start to end, each passage with a timestamp;	FR-001, FR-002, FR-005, SC-001

ID	Role	Flow / click path (URL)	Expected result	FR / SC
			gaps are stated, nothing invented. Speaker labels shown where the data has them (verify what the QA chapter carries).	
QA-001-07	student	Play the recording on the transcript page.	The spoken passage is highlighted, the highlight moves, and it is scrolled into view; highlight carries `aria-current` and a non-colour marker. "Follow" toggle and "resume" control work.	FR-009, SC-003
QA-001-08	student	Click a passage timestamp. Then open `/chapters/01/transcript?t=120` and a `#p-<pid>` link.	Recording lands on the passage time (within 5 s), plays or says the browser refused, transcript scrolls to the passage. A passage without timestamp is disabled and says why.	FR-002, SC-003
QA-001-09	student		Uncertainty is shown by a	FR-003, FR-052

ID	Role	Flow / click path (URL)	Expected result	FR / SC
		Find an "uncertain" passage in the transcript (marker next to text).	text label or icon with an accessible name, not by colour alone; no invented text fills the gap.	
QA-001-10	student	Transcript page: look for the measured accuracy figure and method for the chapter.	Accuracy statement with method is shown, or "unmeasured" is stated explicitly (verify where the SPA places it).	FR-004, SC-002
QA-001-11	student	Open the transcript of a chapter with no transcript (if the QA corpus has one) or `/chapters/01/transcript#p-AAAAAAAAAAAAAAAAAAAAAAAAAA`.	"No transcript for this chapter" / "unavailable" state, recording still plays; an unknown passage anchor is reported as unresolvable, not a blank page.	FR-049, FR-020
QA-001-12	student	Open `/search`, type a meaning-based phrase sharing no literal words with a known passage.	Results are by meaning; each hit shows kind, chapter, time and a destination.	FR-014, SC-007, SC-008
QA-001-13	student	In `/search` type slowly and watch the suggestion list;	Suggestions appear as you type, are	

ID	Role	Flow / click path (URL)	Expected result	FR / SC
		use Down/Enter on a suggestion.	keyboard operable, and the state "could not answer" is shown as such, not as an empty menu.	FR-015, FR-042, SC-005
QA-001-14	student	Click a search hit.	Lands on the passage (`/passage/<pid>` or transcript at the time) with surrounding context ("Around this passage") and the passage's own timing.	FR-017
QA-001-15	student	On a passage, open the cross-references.	Related material elsewhere in the curriculum is listed; none found shows an explicit empty state.	FR-018
QA-001-16	student	Search `/search` for nonsense text such as "zzqxv plorf".	Plain "no relevant match" with near-misses labelled and a floor caveat; unrelated results are not presented as answers.	FR-019
QA-001-17	student	Operator stops the semantic leg or index (or use a build	Search says unavailable or shows a persistent	FR-020, FR-025

ID	Role	Flow / click path (URL)	Expected result	FR / SC
		where it is down), then search.	"degraded" banner beside real results; never an empty result list.	
QA-001-18	student	Open `/ask`, type a question, submit.	With answering OFF: an "answering is off" banner/ state, no answer, `/search` and browsing unaffected. (Answers with citations / declines are not testable on this instance.)	FR-025 (FR-021, FR-022 not exercisable here)
QA-001-19	student	Tab through `/`, `/chapters`, `/search` from page load; check skip link, focus ring, and a dark/light toggle if present.	Skip link first and moves focus to main; every control reachable; focus always visible; no axe A/AA violations.	FR-041, FR-042, SC-017
QA-001-20	student	Open `/status`.	Platform status measured from the backend (not asserted); `/api/health` agrees with what `/` serves.	FR-012
QA-001-21	student		Explicit "Not found" page	FR-020

ID	Role	Flow / click path (URL)	Expected result	FR / SC
		Open `/chapters/does-not-exist`, `/nowhere`.	(or typed chapter-absent state), not a silent landing on the home page.	
QA-001-22	student	Reload `/chapters/01/transcript` mid-way, then open `/progress`.	Position/ progress is preserved or its absence is stated (verify what the SPA keeps for chapter position).	FR-010
QA-001-23	student	Sign in as student1 and try `/admin/users`.	Student sees the student side only; admin routes refused.	FR-051
QA-001-24	professor-admin	Sign in as milosvasic, open `/admin/users`.	Role shown; administration and account list visible.	FR-051
QA-001-25	operator	`scripts/status.sh` then `scripts/stop.sh` / `scripts/start.sh` (documented in docs/quickstart.md and docs/manual.md).	Documented start/stop/ status commands work with no undocumented setup; status distinguishes up/down.	FR-012, SC-004
QA-001-26	operator	Confirm docs exist and are honest: `docs/quickstart.md`, `docs/user-guide.md`, `docs/manual.md`, `docs/faq.md`, `docs/limits.md`.	Files present, state what the system cannot do (limits).	FR-030, FR-031
QA-001-27	operator	Run `scripts/add-chapter.sh` twice on a fixture chapter;	Second run duplicates	FR-026, FR-027,

ID	Role	Flow / click path (URL)	Expected result	FR / SC
		run it with materials missing; run two at once.	nothing; missing materials reported precisely and chapter not published; concurrent run refused.	FR-028, FR-054
QA-001-28	operator	`scripts/redact.sh` on a test passage; then search/ transcript for it.	Passage suppressed everywhere (transcript shows a redaction notice); redaction step is required before export.	FR-039, FR-038
QA-001-29	operator	`GET /api/passages/<pid>` before and after a transcript correction + reindex (needs a correction tool run).	Passage id unchanged; text marked machine vs human-corrected.	FR-037, FR-038, SC-016

Operator-only / pipeline requirements not exercisable in the UI: FR-007, FR-056, FR-057 (reassembly/checksum), FR-016/FR-043 to FR-048 (indexing, reusable components, public repos), FR-029, FR-032 to FR-036, FR-040, FR-050, SC-012 to SC-015, SC-018 to SC-023.

Spec 002 — Knowledge areas and deep linking

Adds a knowledge model: areas, terms, lesson sections and questions derived from the corpus, each evidenced by passages, with stable ids and one resolver so every link (area, term, lesson, question, passage, media moment) can be followed both ways. Extends search to those kinds, adds practice question sets, area exports, and OCR on-screen text as an evidence modality. Routes in the candidate: `/` (area grid), `/areas/:slug`, `/practice/:slug`, `/term/:term`, `/passage/:pid`, `/`

`section/:section`, `/search`, `/progress`. There is no SPA view for area export, per-area evidence, coverage, or graph traversal; those are API-only (rows marked).

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
QA-002-01	student	Open `/`, scroll to "Knowledge areas".	Area cards (`area-card-<slug>`) with title, summary and tags; no card shows an opaque id as title. If the API lacks areas, an honest "not served by this build" state with the recording sections listed.	FR-001, FR-006, FR-040	TESTED-BY-AU e2e/area-lesso identifies itsel platform/gate
QA-002-02	student	Click an area card, `/areas/<slug>`.	Breadcrumb "Curriculum > area", title, summary, tags (each a link into `/search`), read minutes, chapter tags linking to `/chapters/<id>`, corpus-revision line.	FR-007, FR-040, SC-001	TESTED-BY-AU src/app/featur revision.spec. e2e/area-lesso run on mocke
QA-002-03	student	On `/areas/<slug>` read "Where it is taught"; click the bracket on the rail and each jump in the jump list (multi-span areas).	Opens the transcript at that exact moment (`/chapters/<id>/transcript?t=...`); each authored span has its own jump.	FR-019, FR-021, SC-009	TESTED-BY-AU e2e/lesson-tim frontend/e2e/ time gramma DEFECT: deep `precision-seg` defects-regist
QA-002-04	student	On `/areas/<slug>` scroll to "Lessons"; expand, tick	Lessons render in order with materials; completion persists	FR-014, FR-017, FR-039	TESTED-BY-AU e2e/area-lesso surface"); plat verify.spec.ts

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		lessons done, reload.	after reload; prev/ next work. Authored text and quoted/transcribed passages are visually distinct.		frontend/src/a quote-distinct
QA-002-05	student	Lesson video material: click its time link.	Opens the RIGHT chapter at the RIGHT time; a document material shows its resolved place, or a stated unresolved note.	FR-019, FR-021, SC-009	TESTED-BY-AU e2e/area-lesso material carri RIGHT time"); lessons_locati
QA-002-06	student	`/areas/<slug>`, "Test" section before completing lessons, then after.	Locked state names the missing lessons and shows no questions; after completing, the test is takeable and shows the pass mark.	FR-034a, FR-039 (FR-010 of spec 004 for the gate)	TESTED-BY-AU e2e/area-lesso before the les once every re
QA-002-07	student	Submit the test with wrong and free-text answers.	Score shown; wrong answer returns the expected answer and explanation; free text makes the percentage a lower bound; assembled model answers are marked "model answer".	FR-038, FR-009 (spec 004)	TESTED-BY-AU e2e/area-lesso and a free-tex comes back w
QA-002-08	student	Open `/practice/<slug>`.	Question kinds (multiple choice, flashcard, short answer) with a kind badge; each shows its citations; answers are recorded with an acknowledgement	FR-034, FR-034a, FR-035, FR-039	TESTED-BY-AU e2e/016-live-v answers recor adaptive-prac internal/api/p

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
			and survive reload. Short and long question sets both present (verify how the SPA labels them).		
QA-002-09	student	On a practice question, follow its citation.	Lands at the passage / media position named by the citation; a question with an unresolvable citation is not served (shows "citations withheld"/"uncited" note instead of the question as cited).	FR-035, FR-036, FR-037, SC-018, SC-019	TESTED-BY-AU pkg/assessme (`TestSC019_Qu` platform/gate partial: the UI end spec
QA-002-10	student	Open `/term/<term>` for a term from an area or a search hit.	Term article with "Where this term appears" occurrences; unknown term shows "There is no term by that name."; withdrawn term shows withheld state.	FR-008, FR-031	TESTED-BY-AU internal/api/p API outcomes browser: no e was found. KN minted passag platform/gate
QA-002-11	student	Open `/passage/<pid>` for a doc_section or code hit.	Passage text with source label and locus ("Where"), neighbours ("Around this passage"), copy-link control; unknown pid shows "absent", withheld pid offers no copy control.	FR-020a, FR-029, FR-030	TESTED-BY-AU internal/api/p (`TestPassageF` platform/fron (passage copy
QA-002-12	student	On a transcript	Lists the areas / lesson section it		TESTED-BY-AU e2e/passage-k

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		passage, open its knowledge disclosure ("where does this appear").	belongs to as real in-app links; unattached passages say why; redacted vs not-in-registry shown distinctly in the component.	FR-019, FR-024a, SC-015a	link to area, n backend/inter knowledge_re redacted vs no applied, `learn` platform/gate
QA-002-13	student	`/search`: type an area or term name; read suggestions.	Suggestions include area and term names; choosing one navigates to it.	FR-031	TESTED-BY-AU internal/api/su suggest_target browser spec human verify
QA-002-14	student	`/search`: use the kind filter checkboxes and the area filter select.	Hits narrow by kind and area; the filter is echoed on OK, no-match and unavailable; a changed-but-unapplied filter shows "dirty" note.	FR-032, FR-033	TESTED-BY-AU e2e/inquiry-ev are reported", internal/api/se
QA-002-15	student	`/search`: read a hit's locus; click it.	Every hit carries a locus (unit and position) that resolves; a hit whose locus cannot resolve is not shown.	FR-029, FR-030, SC-014	TESTED-BY-AU pkg/search/loc pkg/search/de frontend/e2e/s links exactly v destination")
QA-002-16	student	`/search` states: result, no relevant match, could not determine.	Three states stay distinct, including the semantic-leg-down case.	FR-033	TESTED-BY-AU e2e/search-de states, render
QA-002-17	student	Search for a content kind that is not indexed (e.g. lesson text, or on-screen	A kind is not offered unless indexed. Lesson sections are not advertised; screen_text must	FR-033c, FR-066	NOT-AUTOMA filter list by ey `lesson_sectio` `not-indexed` in registry.tsv

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		text if offered in the kind filter).	not be advertised until accuracy is measured (verify the kind list on this build).		
QA-002-18	student	Open `/progress` and the profile area view (`/profile`).	Per-area and per-question-set progress shown; numbers match what was done in QA-002-04 and QA-002-08.	FR-039	TESTED-BY-AU e2e/profile.sp area detail ve core/api-prog
QA-002-19	student	With reduced-motion enabled in OS/browser, load `/`, `/areas/<slug>`, open a rail.	Motion effects stop but the final state is shown; no information lost.	FR-044a, FR-044b, SC-029b	TESTED-BY-AU e2e/reduced-m e2e/comfort-m prove-motion
QA-002-20	student	Axe/keyboard pass of `/`, `/areas/<slug>`, `/practice/<slug>`, `/term/<term>`, `/passage/<pid>` in light and dark.	Zero A/AA violations; deep links operable by keyboard; styling uses tokens (no literal brand colours).	FR-042, FR-044, SC-021, SC-022	TESTED-BY-AU e2e/design-ev surfaces); plat tokens.spec.ts responsive.sp
QA-002-21	student	Open a deep link for a redacted target and for a nonexistent target.	Learner sees the same neutral "unavailable" outcome for redacted and not-present (no hint redaction exists); an unresolvable link never lands on a different target.	FR-024, FR-024a, SC-011, SC-011a	TESTED-BY-AU verify-g-kg-24 platform/back KNOWN-DEFE `areas/{area}/` `collapse-scop` registry.tsv

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
QA-002-22	professor-admin	`GET /api/areas/<slug>/export` (all four formats) for a published area; also for an unpublished one.	Markdown, HTML, Word, PDF produced from the same Markdown with citations kept; an unreviewed area is not exported and the omission is stated. (No SPA view exists for export.)	FR-045 to FR-049, SC-024, SC-025	TESTED-BY-AU internal/api/e platform/gate
QA-002-23	operator	`GET /api/areas/<slug>/evidence` and `/api/areas/<slug>/coverage` for a published area. (No SPA view.)	Evidence rows carry resolving positions; coverage is per-area and states no threshold; redacted rows omitted with a count.	FR-008, FR-020, SC-001, SC-017c	TESTED-BY-AU internal/api/e backend/inter
QA-002-24	operator	`GET /api/graph/traverse?from=<area pid>&depth=2` (No SPA view.)	Reachable neighbours returned; cycles terminate; dangling hops reported.	FR-018, FR-026, SC-008, SC-015a	TESTED-BY-AU internal/api/g unit fixtures). `registry-sync` registry.tsv (re on the served verify)
QA-002-25	operator	Redact a passage, then check `/search`, `/areas/<slug>`, `/passage/<pid>`, the area export and any cached suggestion.	The redaction is absent from every artefact kind and from cached/ precomputed offsets and snippets.	FR-027, FR-027a, SC-012, SC-012a	TESTED-BY-AU verify-redacti gates/verify-g
QA-002-26	operator				

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		Run `platform/gates/verify-connectivity-matrix.sh` and `platform/gates/verify-search-latency.sh`.	All six traversals non-empty; suggestion p95 under 200 ms and search p95 under 2 s (single client).	FR-033a, SC-015a, SC-016, SC-017	TESTED-BY-AU verify-connec verify-search- under 5-way l `under-concurr` registry.tsv
QA-002-27	operator	Read `docs/on-screen-text.md`; check that no `screen_text` mention is published without a per-chapter OCR accuracy figure.	OCR accuracy published beside speech figures; no OCR-derived mention public for an unmeasured chapter.	FR-060 to FR-066, SC-031 to SC-036	TESTED-BY-AU verify-ocr-eva ocr-redaction. browser flow)

Operator-only / pipeline requirements not exercisable in the UI: FR-004a, FR-005, FR-009 to FR-013 (extraction idempotence and term significance), FR-033d to FR-033j (pipeline), FR-053 to FR-059, SC-002 to SC-007, SC-010, SC-013, SC-026 to SC-030, SC-015b to SC-015g.

Spec 003 — Chapter hierarchy

Makes the chapter id a dotted, zero-padded path (`01`, `02.01`, `02.01.03`) from which parent, depth, ancestors and children are derived, never stored. Adds `hierarchy` to chapter responses and `under`, `depth`, `include_self` filters on `GET /api/chapters`; no new route, no new screen. The SPA shows the ordinal only (chapter card, list, detail eyebrow). Whether the QA corpus contains any sub-chapter (e.g. `02.01`) is UNKNOWN; if it holds only flat chapters, the sub-chapter rows below are API/fixture checks only.

ID	Role	Flow / click path (URL)	Expected result	FR / SC
QA-003-01	student			

ID	Role	Flow / click path (URL)	Expected result	FR / SC
		Open `/chapters` and look at the ordinal on each card, then `/chapters/<id>`.	Ordinal appears (the middle-dot appears only when none was sent); chapter 1 reads the same as before this feature; detail eyebrow reads "Chapter ".	FR-022, FR-023, FR-025, SC-013, SC-022
QA-003-02	student	Open `/chapters/02.01` (only if such a chapter exists on this build; otherwise open a made-up dotted id).	Existing sub-chapter opens through the same page as a top-level chapter; a made-up id shows the chapter-absent / not-found state, not the home page.	FR-026, FR-032, SC-015, SC-023
QA-003-03	student	In `/chapters` and `/` verify order of chapters including any dotted ids (e.g. `02`, `02.01`, `02.10`, `03`).	Parent immediately before its first child; `02.09` before `02.10`; same order in list and curriculum page.	FR-009, FR-010, SC-003, SC-004
QA-003-04	professor-admin	`GET /api/chapters` with a token; inspect one row.	Row has `hierarchy` with parent_id, depth, ancestor_ids (root first), child_ids, ordinal_path; top-level row has parent_id null; leaf has child_ids `[]`; `chapters` is a flat array.	FR-006, FR-008, FR-027, FR-039, SC-016, SC-029
QA-003-05	professor-admin	`GET /api/chapters?under=02`, `?under=02&depth=1`, `?under=02&include_self=true`.	Only descendants (depth=1 gives direct children only); include_self adds the parent; the filters are echoed in the response on every status.	FR-028, FR-040, SC-017, SC-030
QA-003-06	professor-admin		Both return 200, empty `chapters`,	FR-029, FR-041,

ID	Role	Flow / click path (URL)	Expected result	FR / SC
		`GET /api/chapters?under=99` and `?under=2.2` (grammar-invalid).	`under_resolved:false`, same shape; never 404.	SC-018, SC-031
QA-003-07	professor-admin	`GET /api/chapters/<id>` for a sub-chapter whose parent directory is missing (orphan), if the QA data has one.	Served with `orphaned:true` and the missing ancestor named; chapter list still includes it.	FR-031, FR-042, SC-019
QA-003-08	professor-admin	`GET /api/chapters` over an unreadable tree vs. an empty `chapters/` directory (operator-created state only).	Empty readable tree: 200 with `chapters:[]`. Unreadable tree: 503 distinct from empty.	FR-038, SC-028
QA-003-09	student	Search for text spoken in chapter `01` and in a sub-chapter; open hits.	A hit scoped to chapter `01` opens chapter `01`, never the sub-chapter, and the redaction/ingest scope of `01` excludes `02.01`'s artefacts.	FR-012, FR-013, SC-007, SC-008, SC-010
QA-003-10	operator	Run `platform/gates/verify-chapter-hierarchy.sh`, `verify-chapter-depth-unbounded.sh`, `verify-no-hierarchy-fields.sh`.	Exit 0 each (2 means could-not-determine, not a pass). Order identical across API, glob and byte order; no depth limit; no stored hierarchy field.	FR-003, FR-007, FR-009, FR-032, FR-034, SC-003, SC-011, SC-020, SC-021
QA-003-11	operator	List chapter directories under the chapters tree; create (on a scratch copy only) one named `2.1` or `02.2`.	The validator rejects non-zero-padded names and reports them by name rather than skipping them.	FR-004, FR-005, FR-015, SC-001, SC-002
QA-003-12	operator	Review the per-request cost of hierarchy fields (`GET /api/chapters` timing before/after).	The measured cost is published, not assumed.	FR-043, SC-032

Operator-only / pipeline requirements not exercisable in the UI: FR-014 to FR-021 (pipeline patterns, `ingest.sh` defaults, build scripts), FR-033 to FR-037 (route manifest and gate rules), SC-005, SC-006, SC-009, SC-024 to SC-027.

Spec 004 — Authored AI areas

Makes the area catalogue a set of authored AI/IT subjects (title, summary, tags, ordered lessons with materials, video anchors, an end-of-area test with a pass mark) in place of machine-derived entries, with a single publication decision: an area is listed if and only if its detail, lessons and assessment are all servable. Everything is reached through `/` (catalogue), `/areas/<slug>` (detail, lessons, test) and `/practice/<slug>`; there is no separate route per lesson or per test.

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
QA-004-01	student	`/`, "Knowledge areas": open every card you can (or a sample of 10+).	Every listed area opens (zero not-found responses); no card or page title is an opaque id; each shows title, summary, tag(s).	FR-001, FR-003, FR-018, SC-001, SC-003, SC-005	TESTED-BY-AU consistency.sh backend/inter (`TestAreasHan` platform/back (`TestAreaList`
QA-004-02	student	Read the area list: do all areas look like AI / IT subjects, each once?	Only AI/IT subjects; no duplicate subject.	FR-002, FR-020, SC-002	NOT-AUTOMA classification) not enforced, gates/defects-r topicality-clas
QA-004-03	student	On `/areas/<slug>` check the descriptive fields (tags, read minutes, chapters, evidence volume).	Listing fields survive to the page; evidence volume reported, including for thin areas; no area shows no	FR-004, FR-005b, FR-019, SC-016	TESTED-BY-AU areas_test.go (partial: UI dis verify)

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
			evidence figure.		
QA-004-04	student	`/areas/<slug>`, Lessons: read the order, titles, estimated reading time, prev/next.	At least one lesson; defined order; each has title, body, reading time computed from body; neighbours reachable without returning to the area.	FR-006, FR-007, FR-007a, SC-004	TESTED-BY-AU area-lesson-pr href.spec.ts (u
QA-004-05	student	In a lesson with materials: check each illustration/ diagram/video has a caption and text alternative.	Caption and alt text present; a material with no alt text says so (`material-noalt`).	FR-013	TESTED-BY-AU platform/fron distinguished
QA-004-06	student	Click a video anchor in a lesson; then do it by keyboard only (Tab, Enter).	Recording opens at the referenced start, transcript scrolls to the referenced passage and visually marks it, arrival is announced to assistive tech (live region), within 1 s of the moment.	FR-014, FR-015, FR-015a, SC-008	TESTED-BY-AU keyboard.spec frontend/e2e/ transcript-anc
QA-004-07	student	Follow an anchor whose passage/ recording is unavailable (operator seeds a bad ref on a	Learner is told the target is unavailable; no empty/not-found page.	FR-016, FR-017	TESTED-BY-AU keyboard.spec

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		scratch build, or use `#p-<unknown>` on a transcript).			
QA-004-08	student	In a lesson that includes a quoted/ transcribed passage, compare it with the authored prose.	Quoted text is visually and structurally distinct (quote element, label), not mistakable for authored prose.	FR-016b, SC-020	TESTED-BY-AU area-lesson-qu
QA-004-09	student	`/areas/<slug>` Test section on an area WITHOUT completed lessons; also inspect the network response of the assessment call (DevTools).	Test is locked and names the remaining lessons; response contains no question text, options or answers.	FR-010, FR-012a, SC-009	TESTED-BY-AU test.spec.ts ("L backend/inter (`TestGradedTe` graded-test-w
QA-004-10	student	After completing lessons, open the Test: read the pass mark before the first answer; take the test; submit.	Pass mark shown before attempt; score with per-question correct/ incorrect and the correct answer; correctness shown by icon or label, not colour alone.	FR-008b, FR-009, FR-022a, SC-006, SC-017	TESTED-BY-AU test.spec.ts ("T correctness-in
QA-004-11	student	Re-open the same test result after submitting (reload; open in a second tab).	The same result is returned each time.	FR-011	TESTED-BY-AU assessment_re (`TestAssessme`
QA-004-12	student	Submit a blank or partially filled	Only the attempted	FR-009, FR-012	TESTED-BY-AU assessment_d

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		test and inspect the response.	questions' answers are disclosed; a blank submission discloses no answer key.		(`TestBlankSub` `TestPartialSu` platform/gate
QA-004-13	student	In a test with a bank whose size makes the threshold non-integral, check the "pass mark" and required count.	Required correct count rounds up, never down.	FR-008c	TESTED-BY-AU lessons_test.g (`TestAssessme` `TestLearnerAt`
QA-004-14	student	Open an area that has no question bank.	Area page says "No test is being served for this area" with the reason; practice deck is still offered; the test section is not an error.	FR-008, FR-010	TESTED-BY-AU area-detail-no lessons-and-te areas per SC-0
QA-004-15	student	Full learner loop: `/` then an area, finish its lessons, take the test, see the score, with no not-found at any step.	Whole journey completes inside the product.	SC-006	TESTED-BY-AU test.spec.ts (pa platform/gate
QA-004-16	student	Scan the app for a page that appears to wait without saying what it is doing (loading an area, submitting a test).	Every wait shows a named loading/ working state.	SC-015	NOT-AUTOMA must look
QA-004-17	student				

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		Look at the colour of the principal surfaces (home, area, practice) in light and dark.	Six or more distinct hue families; text contrast OK in both themes.	FR-021, FR-022, SC-010, SC-011	TESTED-BY-AU platform/gate design-tokens
QA-004-18	student	Check the area page footer / catalogue response for the corpus revision (`corpus-revision` line; or `corpus_revision` in `GET /api/areas`).	Corpus revision is declared on the catalogue and each area.	FR-023a, SC-018	TESTED-BY-AU corpus_revisi area-detail-co identity.sh
QA-004-19	professor-admin	`GET /api/areas?include=held_back` with a token.	Withheld areas keep their identifier, reason and (where derivable) a display name; an inconsistent build is distinguishable from an editorial refusal.	FR-018b, FR-018c	TESTED-BY-AU areas_test.go platform/back (`TestHeldBack`
QA-004-20	professor-admin	Open an unpublished area by direct URL `/areas/<held-back-slug>` and try `GET /api/areas/<slug>/export`.	Detail identifies itself without leaking unpublished lesson/ question text; export is refused.	FR-018e, SC-019	TESTED-BY-AU export_public passages_pub platform/back
QA-004-21	operator	Run `platform/gates/verify-placeholder-sweep.sh` and	No placeholder/ TODO/stub-	FR-012b, FR-012c	TESTED-BY-AU sweep.sh; plat

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		`verify-lesson-content.sh`.	shaped lessons are published.		
QA-004-22	operator	Run `platform/gates/verify-material-metadata-disclosure.sh`.	No EXIF location, local path, or host identifier in published materials.	FR-013a	TESTED-BY-AU disclosure.sh
QA-004-23	operator	Grep the authored lessons/ questions for names of non-owner participants (use the project's content-boundary gate).	No private individual is named, quoted or identified.	FR-005c	NOT-AUTOMA is `scripts/ver` design per pro required
QA-004-24	operator	Time `GET /api/areas` and a reference resolution at current scale; run `platform/gates/verify-catalogue-scale-latency.sh`.	Latency measured and reported, not assumed.	FR-020a	TESTED-BY-AU latency.sh; pla catalogue_sca
QA-004-25	operator	Clone a learning module into an empty directory outside the umbrella and run its documented bootstrap.	Module reaches its self-reported ready state.	FR-024, FR-025, SC-012	NOT-AUTOMA `standalone.sh` was found

Operator-only / pipeline requirements not exercisable in the UI: FR-026 to FR-030 (check discipline, non-empty-set rule, corpus stability), FR-008a/FR-008 question authoring, FR-018a (single decision point, code structure), SC-013, SC-014, SC-004a.

Findings for the caller

- Row counts: Spec 001 = 29, Spec 002 = 27, Spec 003 = 12, Spec 004 = 25 (93 total).
- No SPA screen exists in the candidate for: area export (`GET /api/areas/{area}/export`), per-area evidence and coverage endpoints, graph traversal (`/api/graph/traverse`), and the chapter-hierarchy filters (`under`, `depth`, `include_self`) or the `hierarchy` object (spec 003 has no UI by design, FR-026). These rows are API/operator checks.
- No browser e2e spec was found for the term page (`/term/<term>`, `term.component.ts`) or for a sub-chapter page; `/section/:section` is routed but no flow in these four specs targets it (its content was not inspected).
- Not exercisable on this QA instance: cited answers and declines (answering OFF), OCR-derived screen text (operator pipeline), and any sub-chapter flow unless the QA corpus actually contains a dotted chapter id.
- KNOWN-DEFECT evidence comes only from `platform/gates/defects-registry.tsv` (dated entries); no file under `the review reports (by file name)` addresses specs 001 to 004.

Part B

Candidate tree: `~/Projects/vasic-wt/qa-candidate` (workshop; abbreviated `W/` below). Umbrella: `~/Projects/vasic` (`U/`). Specs: `U/specs/<NNN>/spec.md`. Review sources abbreviated: `E4` = `~/.cache/claude-tmp/reviews/parts/part-E-sub4-005-009.md`, `E5` = `~/.cache/claude-tmp/reviews/parts/part-E-sub5-008-010.md`.

Reading notes

- Specs 005, 006 and 007 are SUPERSEDED by 008 (each spec.md says so). Their requirements are still listed per spec because the FR/SC ids are what the tasks cite.
- SPA routes verified in `W/platform/frontend/src/app/app.routes.ts` and feature route files: `/login`, `/` (curriculum), `/chapters`, `/chapters/:slug`, `/chapters/:slug/transcript`, `/areas/:slug`, `/practice/:slug`, `/ask`, `/search`, `/progress`, `/plan`, `/passage/:pid`, `/term/:term`, `/section/:section`, `/documents/:slug/read`, `/status`, `/textbooks`, `/library`, `/materials`, `/profile`, `/profile/medium`, `/settings/providers`, `/account`, `/admin/users`, wildcard not-found. There is NO route `/areas` (list) or `/sub-chapters`; the area list is a section of `/`.
- The four session-record sections (meeting notes, open questions, TODO, next meeting) have NO own SPA route. They render as four

panels on `/chapters/<slug>` and are served on four API routes (`GET /api/chapters/{chapter}/open-questions|todo|next-meeting|meeting-notes`, `W/platform/backend/internal/api/session_sections.go`).
- Chapters on disk: 01, 02, 02.01, 02.02, 03, 04 (`W/chapters`). Use `/chapters` to see which the QA instance actually lists. "verify" = the spec requires it but I did not confirm the exact UI wording.
- This QA instance has answering OFF and hosted providers OFF. Accounts: administrators milosvasic and rami; students student1..student5. No credentials are written here.
- Automated e2e specs under `W/platform/frontend/e2e/` need a running backend; I cite them as evidence that a check exists, not as a run result from this session.

Spec 005 — A repository that clones, and a fleet with no unexplained red

Delivers an obtainable repository (documented clone path, cost warning) and instruments whose red output states defect / declared / could-not-run. Superseded by 008; mostly operator-checkable, plus three learner-facing items (no-assessment disclosure, withheld-citation questions, palette/contrast).

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
QA-005-01	operator	`bash W/scripts/verify-obtainability.sh`	Prints the obtaining cost and a verdict. rc 0 = below warning threshold, rc 1 = WARN/BREACH, rc 2 = could not ask (never a pass). Cost under 250 MB for the documented path.	FR-001, FR-002, FR-004, SC-002	TESTED-BY AUTOMATI partial: `W/s verify-obtainabil` measures c only; SC-00 clones on a machine th never held repo) is a s host trial, r by any scri T011 LIVE-
QA-005-02	operator	Read `W/README.md`	States the clone	FR-003, SC-003	KNOWN-DI E4 T012/W

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		section "Obtain it"	command, what a reader gets and what they may not; says recordings are fetched by a documented step.		recorded-n fetch step i built; W/.gitattr is absent (confirmed README sa SC-003 unn
QA-005-03	operator	`bash W/scripts/verify-recorded-material-boundary.sh`	rc 2 (could not determine) while nothing points at external storage; never rc 0 in that state.	FR-024, SC-016	TESTED-BY AUTOMATI partial: scr exists with proof; E4 V it is not reg in any chec registry or verify.sh, s nothing ru automatica
QA-005-04	operator	`bash W/scripts/verify-transfer-retryability.sh`	Verdict per part retryability; rc 0/1/2 convention.	FR-025	TESTED-BY AUTOMATI partial: W/s verify-tra retryabili proof is a s local fixtur the real sto backend (E
QA-005-05	operator	`bash U/scripts/fleet-summary.sh`	One summary with defect count and declared-condition count shown separately; clean instruments still listed at 0.	FR-006, FR-009, SC-005, SC-008	TESTED-BY AUTOMATI partial: U/s fleet-summ (registered T031); E4 n SC-008 reve DEFECT wh content-bo declared rc (next_revie

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
					2026-10-15 expired
QA-005-06	operator	`bash U/scripts/record-reopen.sh` on a closed red, then read `U/docs/findings/reopened-reds.jsonl`	The reopen is a new recorded line with the contradicting evidence and a timestamp; the original closure is not overwritten.	FR-026, FR-027, SC-017	TESTED-BY AUTOMATI partial: `U/s record-reo` exists; E4 T T059 DONE MERGED (c review, not
QA-005-07	student	`/` then open an area that has no question bank (follow an area card to `/areas/<slug>`)	The overview itself says no test is being served and why (wording: "no test is being served"). Never a silent absence.	FR-013, SC-010	TESTED-BY AUTOMATI `platform/f src/app/fe areas/area no-assessment` (unit, checl text). E4 T0 reason text generic, do split struct corpus-unsupporte
QA-005-08	student	`/areas/<slug>` of an area with a test; open its test	Every served question carries a resolving citation; none is blank or missing. A question whose cited material is withheld is absent, not shown without provenance.	FR-012, SC-009	TESTED-BY AUTOMATI partial: `W/ evidence/s question-b withholdin result.jso dispositio` (source and verified pe T037); the c screen resu a human lo

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
QA-005-09	student	`/` and `/chapters`, in light and dark presentation (verify where the toggle is)	Several distinct hue families on principal surfaces, each with a stated job (e.g. kind chips); text readable in both presentations.	FR-016..FR-019, SC-012, SC-013	TESTED-BY AUTOMATI partial: `W/ platform/g verify-ser palette.sh served-pal jobs.sh`, `ve served-con` measure th served pag T046/T049 run agains final build new surfac changed th population
QA-005-10	operator	`bash U/scripts/verify-check-registry.sh`	rc 0; every registered check has a paired proof; an unregistered new script fails.	FR-020, FR-021, SC-014	TESTED-BY AUTOMATI partial: `U/s verify-che registry.s` E4 T054: th no 005-FR t matrix, so S not shown end

Spec 006 — A complete session record per chapter, and a platform a QA team can test

Delivers four cumulative sections per chapter (meeting notes, open questions, TODO, next meeting) with origin session, carry-forward marker and slip count; lessons, tests and materials in every area; and the QA documents. Superseded by 008.

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
QA-006-01	student	`/chapters` then open each	Page shows four panels:		

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		chapter to `/chapters/<slug>`	Next meeting, Open questions, Meeting notes, TODO. Each has content for that session, or states why it is empty.	FR-001, FR-004, SC-001	TESTED-BY `session_se` (API); pane
QA-006-02	student	Open a later chapter (e.g. `/chapters/02.01`) and compare with an earlier one	Items raised earlier and still unresolved appear in the later record (cumulative).	FR-002	TESTED-BY `platform/b` verify by e
QA-006-03	student	In each panel read the item meta line	Every item names the session it originated in ("Raised in session NN").	FR-003, SC-002	TESTED-BY `forward.sp`
QA-006-04	student	Find a carried-forward point on a chapter that has one	A badge with a circling-arrow glyph and the words "Carried from session NN · slipped N time(s)". Not colour-only. Slip count visible.	FR-009, FR-010, SC-004	TESTED-BY `forward.sp` on main/in on the live
QA-006-05	student	Open the latest chapter, look at "Next meeting"	Carry-forward from the latest chapter lands here even though no	FR-011	NOT-AUTO

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
			later chapter exists.		
QA-006-06	student	Open `/chapters/02.02`, `/chapters/03`, `/chapters/04`	Sections show an honest "no plan recorded" style reason, not a blank. Covered/uncovered status is undecided until the operator confirms verdicts.	FR-006, FR-007, FR-007a, FR-008	KNOWN-D in `W/curri`
QA-006-07	operator	With a token: `GET /api/chapters/<slug>/open-questions`, `/todo`, `/next-meeting`, `/meeting-notes`	HTTP 200, one envelope shape, same items as `session_record` on `GET /api/chapters/<slug>`. Anonymous gets 401. Unknown chapter is a determined negative, not a missing route.	FR-005, FR-012	TESTED-BY `session_se` `TestGateSR` `TestGateSR`
QA-006-08	operator	`bash W/platform/gates/verify-session-record.sh`	rc 0 over the real tree; rc 2 if no Go toolchain (never a pass).	FR-007..FR-012, SC-003	TESTED-BY `session-re`
QA-006-09	student	`/` then an area card, then each lesson (`/areas/<slug>`)	Lessons listed in order with title and estimated reading time; each has real	FR-013, SC-005	TESTED-BY ("lessons r

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
			body text; next/previous work.		
QA-006-10	student	Same area, before finishing lessons, look for the test	Test is locked and names what remains; no questions or answers are shown.	FR-014	TESTED-BY ("LOCKED `disclosure`
QA-006-11	student	Mark every lesson done, open the test, answer, submit	Test becomes takeable; one submit gives a score plus per-question right/wrong and the expected answer; a free-text question makes the score a lower bound.	FR-014, SC-005	TESTED-BY ("TAKEABL makes the
QA-006-12	student	Area with no test	Learner is told there is no test and why.	FR-015, SC-006	TESTED-BY
QA-006-13	student	A lesson containing an image or video material	Renders in place with a caption; an `alt` text exists; a missing file says "unavailable", no broken element.	FR-016	KNOWN-D absence (`m` Also E4 T0
QA-006-14	operator	Read `W/docs/qa/MANUAL-TEST-PLAN.md`, `CLIENT-WALKTHROUGH.md`, `KNOWN-LIMITATIONS.md`,	Files exist; every step has an expected result before the action; limitations disclosed in	FR-018..FR-020b, SC-007, SC-008, SC-015	TESTED-BY checks stru the final m

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		`PASS-FAIL-CHECKLIST.md`, `README.md`	advance; documents state the build id they were written against.		
QA-006-15	operator	`bash U/scripts/zero-gap-qa-readiness.sh`	Read-only readiness verdict for starting a manual QA session; rc 2 when the platform is unreachable (blocking precondition, FR-022b).	FR-021, FR-022b	TESTED-BY
QA-006-16	operator	Check that no recorded content or participant name appears in public tracked files	Records refer to private material by location only.	FR-030, FR-030a, SC-013	TESTED-BY per U/CLA pending)

Spec 007 — Decouple modules, add authentication, extract reusables

Delivers a mandatory sign-in in front of every route and API call, role-based access, durable per-user progress, and standalone modules. Superseded by 008. The ai_interviewing module is NOT served by this QA instance, so its restricted-user rules (FR-011..013) are not exercisable here.

ID	Role	Flow / click path (URL)	Expected result	FR / SC
QA-007-01	anonymous	Open `/`, `/chapters`, `/`	Each is redirected to `/login?`	FR-007, SC-003

ID	Role	Flow / click path (URL)	Expected result	FR / SC
		areas/<slug>, /search in a private window	returnUrl=<requested path>; no content flashes first.	
QA-007-02	anonymous	Browser or curl without a token: GET /api/chapters, /api/areas, /api/progress	HTTP 401 on every /api/* route. GET /api/health is the only exception and returns no content or user data.	FR-007, FR-050, SC-003
QA-007-03	professor-admin	/login, sign in as milosvasic, then as rami (separate sessions)	Lands on the requested page (or /); account name and "Administrator" role badge shown; both have equal access.	FR-009, FR-010, SC-004
QA-007-04	student	/login as student1	Signs in; student sees the student side only; no /admin/users entry.	FR-010, FR-054
QA-007-05	anonymous	/login with a wrong password for a real user, then with a username that does not exist	Both show the same generic error message (role=alert), same shape; neither reveals whether the user exists.	FR-049
QA-007-06	anonymous	Submit wrong credentials more than 5 times in a minute from one browser	Further attempts are throttled with a clear message; correct login works again after the window.	FR-048
QA-007-07	student	While signed in click "Log out" (top bar, account-logout) then press Back	Session ends; protected pages redirect to /login; Back does not show cached content.	FR-014
QA-007-08	student	Log out, sign in as another	New session with the new account's	FR-015, SC-005

ID	Role	Flow / click path (URL)	Expected result	FR / SC
		account, check the page	permissions; nothing of the previous account remains. Time under 3 seconds.	
QA-007-09	student	Complete a lesson in an area, log out, log back in, reopen `/progress` and the area	Progress is still there, tied to the account (restored from the server).	FR-018..FR-020, SC-006
QA-007-10	operator	Run the auth hardening Go tests: `go test ./cmd/workshop-server -run 'TestT505|TestT507'` in `W/platform/backend`	Pass: login never reuses a pre-auth session id; every protected route is denied while the auth DB is down (fail closed).	FR-045, FR-047
QA-007-11	operator	Inspect response headers of any `/` response and the session cookie after login (browser dev tools)	Cookie HttpOnly, SameSite=Strict, Secure only over TLS; CSP, X-Frame-Options DENY, Referrer-Policy no-referrer present. Tokens never in URLs or logs.	FR-017, FR-044
QA-007-12	operator	`bash W/platform/gates/verify-spec007-auth-wiring.sh`	rc 0: WORKSHOP_ENV and AUTH_SHARED_SECRET reach the container; rc 2 = could not render compose (never a pass).	FR-043
QA-007-13	operator	Sign in on workshop, take the token the response returns (do not	Other module rejects a token minted for another audience.	FR-046

ID	Role	Flow / click path (URL)	Expected result	FR / SC
		paste it anywhere), present it to the other module		
QA-007-14	operator	Create/inspect a user with no role in a scratch store	Zero access, never admin-equivalent (role column constrained to admin/user).	FR-054
QA-007-15	operator	`bash W/scripts/verify-standalone-clone.sh` and `bash W/scripts/bootstrap-standalone.sh`	rc 0 = nothing requires the umbrella; rc 2 = could not ask. Clone-install-run within 10 minutes.	FR-001, FR-003, FR-005, SC-001
QA-007-16	operator	Check for a login/roles/rotation guide in `W/docs/` (`user-guide`, `faq`, `manual`, `quickstart`)	Documents explain sign-in and roles.	FR-029, FR-030, FR-032, SC-009
QA-007-17	operator	ai_interviewing restricted-user rules (employer data denied, knowledge base allowed)	Not part of this instance.	FR-011..FR-013

Spec 008 — Unified workshop platform

Merges 001-007 into one platform: chapters, transcripts, session record, areas with lessons and tests, search, ask (answering OFF here), deep links, exports, authentication. About 339 FRs, 168 SCs. Only learner/professor-visible flows are listed; pipeline and operator requirements are summarised in one row.

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Statu
QA-008-01	student	Sign in, land on `/`	Curriculum page: heading "An index of a recording.", sections "The recording(s)", "What is indexed", "Knowledge areas" (area grid) and "Chapters" (chapter grid).	US23, US12	TEST `front` `desig`
QA-008-02	student	`/chapters`	Every chapter on disk is listed with title and summary and hierarchy (parent, depth); if none: "No chapters yet", never a blank. If the list cannot load, an "unavailable" state with a reason.	US12, US16	TEST `brows` every `list`
QA-008-03	student	`/chapters/02.01` (a sub-chapter)	Resolves like any chapter: dotted id in breadcrumb, child falls right after its parent in the list.	US6	TEST `chapt` `api/c` G-CH false
QA-008-04	student	`/chapters/<slug>` of a sub-chapter whose parent folder is absent (if one exists on this	Still served; flagged orphaned; parent id still named.	US21	TEST `backe` `chapt` (API)

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Statu
		instance; verify)			
QA-008-05	student	`/chapters/<slug>`, press play on the recording	Player mounted; recording streams (seeking works, Range support); split parts play without reassembly.	US12	TEST `brows` mou
QA-008-06	student	`/chapters/<slug>/transcript`, click a passage, then use keyboard only	Transcript with timestamps; selecting a passage seeks the video; follow mode highlights the current passage; uncertain segments marked; keyboard operable and announced.	US3, US14	TEST `trans` `keybo`
QA-008-07	student	Chapter with a recording but no transcript	Recording still plays; transcript area says it is unavailable, never reads as silence.	US3, US7	TEST `brows` endp plays
QA-008-08	student	In the transcript use the copy-link control on a passage, open the link in a new private context (signed in)	Lands on the same passage; `?t=` and `#p-` links still work; withheld passage offers no copy control.	US13, US14	TEST `momer`
QA-008-09	student	In an area lesson, follow a "recorded moment" reference	Recording page opens positioned at the start time, transcript scrolled to the passage and highlighted; a range shows its extent; a	US14	TEST `lesso` `trans`

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Statu
			missing target says "unavailable".		
QA-008-10	student	Open any transcript passage's knowledge/ cross-reference disclosure; open `/passage/<pid>` and `/term/<term>` from a search result	Related areas, section and questions listed, each one step away; withheld passages stay withheld; a dead link fails loudly with a reason.	US13	TEST `passa` `trans`
QA-008-11	student	`/areas/<slug>` (from the area grid on `/`)	Title made of words (not an id), summary, at least one tag, "Where it is taught"; detail never says the area does not exist.	US10	TEST `area-` route `gates`
QA-008-12	student	Lessons, lock, test, score	See QA-006-09..QA-006-11 (same flow). Re-opening the result shows the same result.	US11	TEST `area-`
QA-008-13	student	`/practice/<slug>` (and adaptive practice from the area)	Practice set with multiple choice, short answer and flashcard kinds; a question with unresolved citation is withheld; assembled model answers are marked as such.	US11	TEST `adapt` `flash`
QA-008-14	student	Area materials: image, video	Image and video render in place with caption and text	US15	KNO empt

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Statu
		segment, and a lesson with a missing file	alternative; the video is bounded to its range; a missing file says unavailable.		`front` `bound`
QA-008-15	student	`/search`, type a query that appears in the corpus; then a nonsense query	Results state their kind and position (time span or text span); selecting one lands on the exact passage; nonsense query says plainly nothing relevant, not irrelevant results.	US20	TEST `live-` "a qu deter
QA-008-16	student	`/search`, type-ahead: type slowly and watch the suggestions	Suggestions (areas and terms included) appear quickly, with no stall.	US20	TEST `live-` conc `verif`
QA-008-17	student	`/search`, filter by area or kind	Only matching results; the filter is visible in the UI.	US20	NOT- end t
QA-008-18	student	`/search` while the search backend is degraded (operator can stop the embedding backend; verify before doing this on the QA instance)	A "could not answer" state, never rendered as "no results".	US20	TEST `searc` ("an deter
QA-008-19	student	`/ask`, type a question (answering is OFF on the QA instance)	Page says answering is unavailable (HTTP 503, reason `no_provider`), while search and browsing keep working.	US22	TEST `front` disca `answe` item.
QA-008-20	student			US11, US29	

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Statu
		`/progress` and `/plan`	Per-area progress and study plan render for the signed-in user; empty state explicit.		TEST `front` verif
QA-008-21	student	`/section/<section>` from a chapter (e.g. a meeting-notes item link)	Section browser opens with the referenced section.	US13	NOT- end t
QA-008-22	student	`/textbooks`, `/library`, `/documents/<slug>/read`	Textbook list (or one of the stated empty/ held-back/ unreadable messages), library list, document reader.	spec-015/008 FR-008	TEST `textb` PLAN
QA-008-23	student	Any mistyped URL, e.g. `/nonsense`	Explicit "Not found" page; not a redirect to `/`.	US2	TEST `brows` `live-` not-f
QA-008-24	student	Tab through `/`, an area page and the test with keyboard only; resize to phone width	No Level A/AA violation; every control reachable; no horizontal scroll.	US19, US23	TEST `a11y-`
QA-008-25	student	Look at on-screen text (OCR) results in search or a transcript	Specified (kind `screen_text`), pipeline unbuilt. Expect NO on-screen-text results to exist.	FR-183..FR-191, SC-103..SC-108	KNO samp `platf` `scree`
QA-008-26	professor-admin	`/admin/users` after signing in as rami or milosvasic	Account list; administrator-only. A student opening it is refused.	US28	TEST `roles`
QA-008-27	professor-admin	`/status`	Platform status table with a probe row per	spec-008 status	TEST `front`

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Statu
			route (e.g. `probe-chapters`), honest states.		`model` e2e
QA-008-28	professor-admin	Export an area (`GET /api/areas/<area>/export`)	Four formats produced; an area failing publication review is not exported and the omission is reported.	US25	TEST `backe` `expor` `expor` verif
QA-008-29	professor-admin	Count areas: `/` area grid	Client sees 39 areas, not 42, until three documents failing publication review are decided.	SC-054/059/162	KNO with oper
QA-008-30	operator	Pipeline, chapter-addition, redaction, decoupling, submodule currency, evidence and governance requirements (approx. 250 FRs: FR-001..FR-191 pipeline/OCR, FR-256..FR-339 evidence/ auth/ decoupling)	Run `bash W/scripts/verify.sh`, `W/scripts/add-chapter.sh` proofs (`prove-add-chapter-*.sh`), `U/scripts/verify-submodule-remote-sync.sh`; every gate is three-valued (0/1/2, 2 never a pass).	US24, US26..US33	TEST unde 262 F re-ve

Spec 009 — PDF-sourced meeting notes with redaction gate

Delivers meeting-notes items (next meeting, open questions, notes, TODO) minted from a chapter's notes PDF when one exists, else from the transcript, with a redaction check on each item. Pipeline spec; the user-visible part is the four panels on the chapter page.

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
QA-009-01	student	`/chapters/01` and read the four session panels	Items come from the chapter's notes PDF; each cites a PDF section (not a transcript segment). Same content on reload.	FR-002, FR-004, SC-001	TESTED-BY-AUTOMATI `pipeline/extract/test_meeting_notes_pd` What the panels show AUTOMATED
QA-009-02	student	`/chapters/02.02` and read the four session panels	Items are present from the transcript; not "unwritten" for all four kinds.	FR-006, SC-004	NOT-AUTOMATED — n exercises it end to end; (E4: "22 items" for 02.0 no `curriculum/chapter` directory)
QA-009-03	student	Scan the panels of every chapter for e-mail addresses, phone numbers, or a participant name	None appear.	FR-007, FR-008, FR-011, SC-005, SC-007	TESTED-BY-AUTOMATI `pipeline/extract/test_meeting_notes_pd` roster name, e-mail, ph of the served text is no
QA-009-04	student	A withheld item (if any panel	Item is not shown; a withheld	FR-007, FR-010	TESTED-BY-AUTOMATI `platform/backend/inte`

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		shows "withheld" for an entry)	notice with a reason (not a silent gap).		`publication_session_t` (`session_evidence_with`
QA-009-05	operator	Re-run the meeting-notes stage twice on the same chapter 01	Byte-identical output; PDF used, transcript not used.	FR-002, FR-005, SC-002	TESTED-BY-AUTOMATI `extract/test_meeting_` (determinism tests)
QA-009-06	operator	Chapter directory with two files matching `* - Notes by *.pdf`	Both processed independently.	FR-001a, SC-008	TESTED-BY-AUTOMATI `extract/test_meeting_`
QA-009-07	operator	Corrupt notes PDF for a chapter	Explicit failure state for that chapter; no fallback to transcript.	FR-009	TESTED-BY-AUTOMATI `extract/test_meeting_`
QA-009-08	operator	Transcript-only chapters (e.g. 02, 03, 04) before/ after	Previously served meeting-notes content unchanged.	FR-003, SC-003, SC-006	TESTED-BY-AUTOMATI `pipeline/extract/test_` `test_meeting_notes_pi` no before/after diff of s

Spec 010 — Zero-gap verified closure

Delivers a machine-derived gap register, three-valued checks with paired proofs, tamper-evident evidence and daily re-measurement, all in the umbrella. No learner-facing UI. All rows are operator commands run in `U/`.

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
QA-010-01	operator	`bash U/scripts/zero-gap-sweep.sh` twice on an unchanged tree, diff the outputs	Byte-identical reports; each defect class states its measured recall; un-inspectable areas listed with reason.	FR-001..FR-005, SC-001, SC-002	TESTED-BY-AUTOMATIO `U/scripts/ze gap-determinism.` (T035; not ru by me)
QA-010-02	operator	`bash U/scripts/zero-gap-determinism.sh`	rc 0: sweep prints byte-identical reports over N repeats on a real faithful copy.	FR-003, SC-001, SC-006	TESTED-BY-AUTOMATIO `U/scripts/ze gap-determinism. zero-gap-determinism-modules.sh`; E T064: five-repeat harne over every registered check is PARTIAL
QA-010-03	operator	`bash U/scripts/zero-gap-verify.sh` against a closed item	Re-runs the recorded check on the pre-fix and post-fix refs: fails on pre-fix, passes on post-fix; a check that passes on both is refused ("hollow").	FR-007, FR-018, SC-004	TESTED-BY-AUTOMATIO `U/scripts/ze gap-verify.s` (T039). Close item count UNCONFIRM (E5 T043..T04
QA-010-04	operator	`bash U/scripts/zero-gap-coverage.sh` and read `U/`	Every owned module x test-kind cell is a runnable check with	FR-010, FR-011, SC-005	KNOWN-DEFECT: E5 T053: 192 gap cells remain (stress-chaos

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		`docs/zero-gap/coverage-measured.tsv`	evidence, a declared gap with reason, or could-not-run.		perf-load 15, security 15, e 15, accessibil 15, ...); T054-T060 MISSIN
QA-010-05	operator	`bash U/scripts/zero-gap-guard-checks.sh` and `zero-gap-third-party-guard.sh`	Guards pass; third-party items are registered as not-ours, not fixed.	FR-013, FR-021, SC-007	TESTED-BY-AUTOMATIO `U/scripts/ze gap-guard-checks.sh`, `U/scripts/zero gap-third-party-guard.` registered in `scripts/chec registry.tsv` per E5
QA-010-06	operator	`bash U/scripts/zero-gap-evidence-chain.sh` and look for `U/docs/zero-gap/anchor.json`	Tamper-evident chain verifies; anchor file exists.	FR-016	KNOWN-DEFECT: E5 T083: `U/docs/zero-gap/anchor.json` does not exis chain head is null in `cycle 1/freeze.jso`
QA-010-07	operator	Look for `U/scripts/zero-gap-daily.sh` and `U/ops/systemd/`	Daily re-measurement job (once a day and on demand).	FR-019, SC-009	KNOWN-DEFECT: E5 T069/T070: b absent (MISSING)
QA-010-08	operator	Look for `U/scripts/zero-gap-scan-records.sh`	Scans programme records for unmeasured assertions, guessing language, credentials.	FR-020, SC-008	KNOWN-DEFECT: E5 T062: script does not exis
QA-010-09	operator			FR-024, SC-010	

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		Read the register summary (`workable-items-vsc gap summary --json`, if installed) and `U/docs/zero-gap/module-page-walkthrough.md`; time yourself finding one module's state	Answer for any module within two minutes; summary counts consistent with the register.		NOT-AUTOMATED no automate check exercis it end to end human must look. (Golden page fixture exists; live register migration sta UNCONFIRM E5 T037)
QA-010-10	operator	`bash U/scripts/verify-claim-ledger.sh --completeness`	Recorded claims re-measured; carriers and counts registered.	FR-020, FR-024	TESTED-BY-AUTOMATIO partial: `U/scripts/veri claim-ledger.sh`; E5 T066 PARTIA
QA-010-11	operator	Follow `U/specs/010-zero-gap-verified-closure/quickstart.md` end to end	Every step runs as written.	SC-001..SC-011	NOT-AUTOMATED no automate check exercis it end to end human must look. (E5 T07 MISSING, fin audit T079 MISSING)

Part C

Candidate tree: `~/Projects/vasic-wt/qa-candidate` (HEAD 7dff042). Paths below are relative to it unless absolute. QA instance: full SPA + API on its own port, answering OFF, hosted providers OFF. Accounts: administrators (professors) milosvasic and rami; students student1..student5 (initial password, forced change at `/account` first). Anonymous visitors get 401 on `/api` and are redirected to `/login?returnUrl=...`. No credential is written here.

Status key: TESTED-BY-AUTOMATION (file cited, "partial" where it uses mocks or skips without a special environment), NOT-AUTOMATED, KNOWN-DEFECT (review file + finding). Review files live in `the review reports (by file name)` . Older audit tables in `partials/` describe int-0.0.3 @35f2ea3 and are stale where the candidate has since merged more branches; every claim below was re-checked against the candidate tree.

Route facts verified in `platform/frontend/src/app/app.routes.ts` and the feature route files: `/`, `/chapters`, `/chapters/:slug`, `/chapters/:slug/transcript`, `/areas/:slug`, `/practice/:slug`, `/ask`, `/search`, `/progress`, `/plan`, `/passage/:pid`, `/term/:term`, `/section/:section`, `/documents/:slug/read`, `/status`, `/textbooks`, `/textbooks/:id`, `/textbooks/:id/page/:n`, `/textbooks/:id/figures`, `/library`, `/profile`, `/profile/medium`, `/settings/providers`, `/account`, `/admin/users`, `/materials`. Wildcard renders a not-found page.

Spec 011 — Multi-provider LLM providers (hosted rows, audit, LAN locality)

Delivers `openai` and `anthropic` as selectable hosted rows with the existing four safeguards, a per-attempt remote-generation audit log, a third `lan` locality with a real OpenAI-compatible LAN row, and (US4) an operator provider switch. The user-visible surface in the browser is the read-only `/settings/providers` page (spec 024 owns the page; 011 owns what it reports). With hosted providers OFF on the QA instance, most 011 behaviour is operator-side or negative-path. US4 (provider switch, FR-011..013, SC-006/007) has no implementation and no UI in the candidate (tasks T022/T023 are BLOCKED on operator decisions).

ID	Role	Flow / click path (URL)	Expected result
QA-011-01	Anonymous	Open `/settings/providers`; also `curl -i <base>/api/answering/overview`	SPA redirects to `/login?returnUrl=%2Fsettings%2Fproviders`; API answers 401 and no provider data leaks
QA-011-02	Student	Sign in as student1 (after forced password change), open `/settings/providers`	Page shows the administrators-only sentence; no audit table, no provider table of admin detail

ID	Role	Flow / click path (URL)	Expected result
QA-011-03	Professor-admin	Sign in as milosvasic, open `/settings/providers`	H1 "Answering & providers"; state card says answering is OFF with a reason sentence; "Search and browsing are unaffected in every state" line; "Running provider" card: Provider, Model, Endpoint, "Where it runs" locality badge, "Remote opt-in: off", Index generation; only a Refresh button, no control that changes anything
QA-011-04	Professor-admin	`/settings/providers`, scroll the provider table	Catalogue lists the rows including `openai`, `anthropic`, `openai_compatible_lan`; each row has a locality badge (local / lan / remote words, not colour only); credential column says set / not set, never a value; no row that can never work is offered as selectable
QA-011-05	Professor-admin	`/settings/providers`, click Refresh; keyboard Tab to Refresh and the table scroll region	Refresh re-reads without a page reload; polite status announcement only on a CHANGE of state; both controls reachable by keyboard
QA-011-06	Professor-admin	Break the status read (operator stops the answering overview endpoint, or fixture) and reopen `/settings/providers`	"unavailable" is its own state with a retry; it is never worded as "off"
QA-011-07	Professor-admin	`/ask` while answering is OFF	Ask page states answering is off and why, and says the same thing the settings page says (the "off, and why" banner is spec 024 T029)
QA-011-08	Operator	`platform/backend/cmd/answer-providers` with the `-names` flag	Selectable-name list is derived from the registry, includes openai and

ID	Role	Flow / click path (URL)	Expected result
		(documented in `docs/providers.md`)	anthropic, contains no `openai_compatible` stub
QA-011-09	Operator	Start a scratch server with `-answer-provider openai` WITHOUT remote opt-in (never port 8087, never with a real key)	Server refuses at boot with an explicit reason; nothing is sent anywhere
QA-011-10	Operator	After a (fake-server) remote generation attempt, read the audit JSONL beside the registry	One record per attempt, nine fields, no prompt text, no answer text, no credential; file mode 0600; boot refuses if the audit sink cannot be written (fail-closed)
QA-011-11	Operator	LAN provider: point `openai_compatible_lan` at a LAN OpenAI-compatible server	Accepted without remote opt-in only when the resolved address is private/LAN; a public address is refused; `local` locality still refused if it resolves non-loopback
QA-011-12	Operator	Try to switch the active provider while the server runs (any route, any UI)	No such mechanism exists in the candidate: record as "FR-011/012/013 not delivered". Expect no switch control on `/settings/providers`

Spec 012 — Document ingestion: textbooks (as users see them)

Delivers textbook ingestion (EPUB phase 1; PDF/FB2/HTML/DOCX, extraction-failure reporting, OCR gating in later phases) into searchable `doc_section` passages with licence gating, idempotent re-

ingest, scope hidden from `/api/chapters`, and `/passage/{pid}` deep links. The browsing UI users see is the spec-020 textbook library (`/textbooks`), which is how ingested books are read.

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
QA-012-01	Student	`/textbooks`	One card per catalogued book: title, licence basis, live section count, a state word; opens the book	FR-011, SC-005	TESTED-BY `platform/f` `textbooks.` SERVER: th both books counts" (n fixture ser "FIXTURE: card per b basis and
QA-012-02	Student	`/textbooks/:id` open a book, pick a section, then `/textbooks/:id/page/:n`	Page text renders in an article, header "book · section · page"; Previous/ Next page links work; keyboard next page works	FR-011, FR-001	TESTED-BY `textbooks.` SERVER: b page text, by keyboa fixture ser
QA-012-03	Student	`/textbooks/:id/page/<page with no registry row>`	Plain "no text for this page" state, not an error page	FR-017 spirit, SC-008	TESTED-BY `textbooks.` catalogued row in the plain "no t (fixture se
QA-012-04	Student	`/textbooks/:id/figures`; use the filter select	Figure tiles with a "N of M figures" count; each tile links to its page	FR-011 (reader)	TESTED-BY partial `pla` `src/app/fe` `textbooks/` `figures.co` (Karma); `textbooks.` FIXTURE a block. No NOT-AUTO end
QA-012-05	Student	`/textbooks` → "Search textbooks" box:	Hit shows citation (book, section,	FR-012, SC-006	TESTED-BY `textbooks.` SERVER: S

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		type a phrase known to be in a book	page); clicking opens that page		-> hit with opens that
QA-012-06	Student	`/search` (or `/ask`), search text unique to a textbook page	Hit appears in the main search with a textbook citation beside it; a hit link lands on `/passage/:pid` or the textbook page	FR-011, FR-012, SC-006	TESTED-BY `textbooks` SERVER: t shows the textbook h `frontend/s` `features/p` `textbook-c`
QA-012-07	Student	Open `/chapters`	No textbook-scope entry (e.g. no `textbook-...` chapter) appears in the chapter list	FR-006, SC-005	TESTED-BY partial Go `backend/in` `textbooks` browser cl NOT-AUTO
QA-012-08	Student	A withheld (410) book or page in `/textbooks/...`	Plain-words custody message, not a generic error; 503 shows "unavailable" with working retry	FR-017 spirit	TESTED-BY `textbooks` "FIXTURE: a custody "FIXTURE: unavailab working re
QA-012-09	Operator	Run `bash scripts/ingest-textbooks.test.sh` (self-contained suite)	All checks pass; licence-less document refused with zero passages; DRM EPUB refused with a reason; spine order honoured; re-ingest idempotent	FR-002, FR-004, FR-005, FR-007, FR-009, FR-010, SC-002, SC-003, SC-004	TESTED-BY `scripts/in` `textbooks` `reviews/n2` records 22 b3b0b1e)
QA-012-10	Operator	Ingest a scanned PDF with the documented `--ocr` option through `scripts/ingest-textbooks.sh`	Documented behaviour: OCR sections marked OCR-derived. Reviewer found the zero-token guard trips on `tesseract`	FR-018, FR-019, SC-009	KNOWN-D `n2-2026-10` (HIGH). `pi` `pdf_ocr.py` `tesseract` 185), so re sign-off

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
			`--version`, so the documented OCR entry point fails		
QA-012-11	Operator	Ingest an HTML book with more than 200 unclosed `<p>`/`<li>`, and one with `<meta charset="hex">`	Should ingest or refuse cleanly with exit 2	FR-010, FR-015	KNOWN-D `n2-2026-10` (nesting > rejects val F3 (non-te crashes, rc candidate `LookupErro` `pipeline/` `html_secti` F3 may be F2 `MAX_DEP`
QA-012-12	Operator	Ingest a DOCX with an inline text box; read the run report	Report should list lost content in `extraction_failures`	FR-017, SC-008	KNOWN-D `n2-2026-10` (silent loss `extractior`

Spec 013 — Adaptive assessment scoring (ability, downgrade, self-rated cards)

Delivers a practice-bank submit endpoint, an immutable response log, a 0-10 ability estimate (Elo with shrinking K, clamped), the downgrade path (3 wrong in a row serves an easier item, optionally a flashcard) and ability/attempt count on `/api/progress`. The browser surface is the adaptive card on `/practice/:slug` and the ability line on `/progress` and `/profile`. Operator rule to test explicitly: a SELF-RATED flashcard must never raise ability.

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
QA-013-01	Student	`/areas/<slug>` → practice, or `/practice/<slug>`; answer the first	Question appears ("Picking a question near your level..." while	FR-001, FR-002, FR-004, FR-016, SC-001,	TESTED-BY-AUTOM `adaptive-practice.s` `api/practice_answe`

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		adaptive question	loading); selecting an option shows it selected; submit shows correct/ incorrect, explanation, "Your level in this area: N.N, K answered", then "Next question"	SC-002, SC-008	
QA-013-02	Student	Answer one question, reload, open `/progress`	Ability score and attempt count appear on the progress surface and agree with the card; first-ever visit with no study is an honest empty state	FR-013, FR-016, SC-008	NOT-AUTOMATED e `internal/api/profi`
QA-013-03	Student	Answer 3 questions WRONG in a row in one area	Next served question is easier than the normal pick; page shows a neutral eased-path note (not an alarm); an "adjusted" notice appears	FR-011, FR-013, SC-004	TESTED-BY-AUTOM `internal/api/quest` `adaptive-practice-`

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
			(toast, see spec 014)		
QA-013-04	Student	After a downgrade, answer ONE question correctly, then continue	Selection returns to the normal nearest-to-ability pick (downgrade is temporary, not a ceiling)	FR-012	TESTED-BY-AUTOM TestServeNext_Dow
QA-013-05	Student	Downgrade in area A, then practise area B	B is not downgraded (streak is per area)	FR-011	TESTED-BY-AUTOM TestServeNext_Dow
QA-013-06	Student	During a downgrade a flashcard may be served: Tab to "Show answer"/flip, press Enter, then rate "I recalled it" / "I did not" with the keyboard only	Card shows the self-rating honesty note ("practice card ... Your own rating does not change your level"); body sent is `recalled`, never `chosen`; result announced once via a polite live region; "Next question" serves the next item and moves focus to it	FR-011, SC-004	TESTED-BY-AUTOM `practice-flashcard`
QA-013-07	Student	Outside a downgrade, practise normally for	A flashcard is NEVER served outside a	FR-011	TESTED-BY-AUTOM TestKindEasing_AFl

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		many questions	downgrade; only multiple choice		
QA-013-08	Student	Operator rule: reach a flashcard (3 wrong first), note the level from `/progress` (or `GET /api/progress ability_score`), flip and rate "I recalled it", repeat 5 times (cards may repeat), re-read the level	Level must be UNCHANGED or lower; it must never rise from self-ratings; item difficulty is not recalibrated from them	013/016 operator rule; FR-004; FAQ §24 "never raises your level"	KNOWN-DEFECT: `re` `recalled:true` raise code `platform/backe` `correct = *body.Re` no self-rated handli claim a neutral outc the rise is invisible i
QA-013-09	Student	Same as -08 but rate "I did not recall it"	Level must not fall below what a wrong MCQ would do; honest note shown; no punishment framing	FR-012 spirit	TESTED-BY-AUTOM TestPracticeAnswer `recalled=false` fres `n1-2026-10-21.md` F4
QA-013-10	Student	Answer every question in a small area's pool until repeats are served; open `/progress`	Stated repeats are labelled; an integrity note may appear. Honest learners should not be accused for server-initiated repeats	FR-014, FR-015	KNOWN-DEFECT: `re` the server's own rep honest learners)

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
QA-013-11	Student	Submit a stale/ replayed answer (use the browser back button after submitting, then submit again)	Refused at the request boundary with a stated reason; no second score	FR-014	TESTED-BY-AUTOM that refuses the ans `scoring/store_test`
QA-013-12	Student	Complete the graded end-of-area assessment (`/areas/<slug>` test)	Entry written to the same response log; score is the server's	FR-002	TESTED-BY-AUTOM block, mocked serve
QA-013-13	Operator	Run `go test ./pkg/scoring/...` in `platform/backend` and check the response log is append-only	Pass; UPDATE/ DELETE on `response_log` rejected by the store	FR-003, SC-005, SC-006	TESTED-BY-AUTOM
QA-013-14	Operator	Recalibration batch with a hot question (many sessions answer one question)	Batch finishes promptly; server stays responsive	FR-010, SC-007	KNOWN-DEFECT: re first-attempt query `scoring/recalibrat`

Spec 014 — Gamification design system (toasts, badges, level-up modal)

Delivers the achievement toast region (queue, hover/focus pause, coalescing), achievement badges, and a native-dialog modal for level-up / path-adjusted, all with supportive downgrade framing and reduced-motion support. Events come from the spec 013 event feed (`/api/progress`). `app.config.ts` now provides `ACHIEVEMENT_EVENT_SOURCE`,

so the older "inert" finding no longer applies; the toast/modal e2e specs still skip (as UNCONFIRMED) when the region is not mounted or the build exposes no `window.ng`.

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
QA-014-01	Student	`/practice/<slug>`; answer enough questions to cross a level (or have an operator seed a level-up)	A toast appears with the headline and a figure that counts up to its exact value; no reload needed	FR-001, SC-001	TESTED-BY-AUTOMATION: partial (mocked feed, real app config) `e2e/adaptive-practice.spec.ts` "a scored answer raises th level-up toast through tl real app config"; `e2e/achievement-toast.spec.ts` (skips if n mounted / needs dev build)
QA-014-02	Student	Trigger a downgrade (3 wrong answers)	An "adjusted" toast appears in the accent/neutral style: not red, not amber, no warning/danger words or icon; wording is supportive (what changed and why)	FR-003, FR-016, SC-002	TESTED-BY-AUTOMATION: `e2e/achievement-toast.spec.ts` "FR-003: a adjusted toast is accent-bordered, not danger/warning/success" (light and dark)
QA-014-03	Student	When the adjusted event carries a study link: Tab to the link in the toast	One keyboard-reachable link; toast does not navigate by itself	FR-022	TESTED-BY-AUTOMATION: `e2e/achievement-toast.spec.ts` "FR-022: a adjusted toast with a study href offers one keyboard-reachable linl

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
QA-014-04	Student	Hover the pointer over a toast, then Tab focus into it, then leave	Auto-dismiss is paused while hovered or focused and resumes after both end	FR-006, SC-004	TESTED-BY-AUTOMATION: partial `e2e/achievement-toast.spec.ts` "FR-006: hover keeps the toast pa the auto-dismiss time" (hover only; keyboard-focus path is NOT-AUTOMATED in e2e)
QA-014-05	Student	Cause a burst of several events (answer fast across areas / operator seeds five events)	At most two toasts visible, the rest queue; same-kind events coalesce; downgrade events coalesce more strongly	FR-007, FR-008, FR-009, SC-006, SC-007	TESTED-BY-AUTOMATION: partial `e2e/achievement-toast.spec.ts` "FR-007: a burst of five distinct events shows at most tw toasts" (cap 2 per test; coalescing window valu are spec NEEDS CLARIFICATION: verify eye)
QA-014-06	Student	Reach a level-up	A native modal dialog opens: focus on "Continue", Tab/ Shift+Tab stay inside, background inert, Escape closes and returns focus; downgrade ("adjusted") stays a toast, not a modal	FR-014, FR-015	TESTED-BY-AUTOMATION: `e2e/achievement-modal.spec.ts` (FR-014/ O-3, FR-015 tests); skips host not mounted or no `window.ng`

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
QA-014-07	Student	Several modal-worthy events at once; then dismiss one and reload	Modals never stack (one after another); level-up and chapter-complete of the same area merge into one dialog; a dismissed event never reappears, even after reload; a toast is held, not lost, while the modal is open	FR-014	TESTED-BY-AUTOMATION: `e2e/achievement-modal.spec.ts` (never stacks, ONE dialog, acknowledge once, held not lost)
QA-014-08	Student	Browser setting "reduce motion"; trigger toast and modal	No entrance animation; numeric figure shows its final value at once; level-up and adjusted entrances are identical	FR-013, FR-017, SC-008	TESTED-BY-AUTOMATION: `e2e/achievement-toast.spec.ts` FR-013, `e2e/achievement-modal.spec.ts` FR-017/ FR-013, `e2e/reduced-motion.spec.ts`
QA-014-09	Student	Screen reader on (VoiceOver/ NVDA/Orca): trigger a	Text equivalent announced; adjusted announces	FR-004, FR-005, FR-021, SC-003	NOT-AUTOMATED (axe checks exist in `e2e/achievement-toast.spec.ts`/ `achievement-`

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		toast and an adjusted modal	politely, never assertive		`modal.spec.ts`; the real-screen-reader check T03 is operator-gated)
QA-014-10	Student	Open an area page / chapter page with earned achievements	Badges render with an accessible name and a distinct seen/ unseen state; class namespace is not the provenance `.pf-badge` family	FR-010, FR-011, FR-019	TESTED-BY-AUTOMATION: partial (Karma) `platform/frontend/src/app/features/gamification/achievement-badge.component.spec.t` `area-badges-surface.spec.ts`, `chapte` `badges.component.spec.`
QA-014-11	Student	Brand-new account, no activity; open `/progress`, `/profile`	No toast, badge or modal for a null/ loading/ unavailable score; no default "0" level shown as if earned	FR-012, SC-010	TESTED-BY-AUTOMATION: partial Karma `achievement-toast-region.component.spec.` `profile-completeness.spec.ts`
QA-014-12	Student	Phone width (360 px) and an RTL language, trigger a toast and a modal	Both stay inside the viewport; Continue is a comfortable target; no sideways page scroll	FR-013 spirit	TESTED-BY-AUTOMATION: `e2e/achievement-modal.spec.ts` "responsive" and "RTL"; `e2e/achievement-toast.spec.ts` "stays inside the viewport"
QA-014-13	Operator	Independent design-toolkit QA test-bank review of the	Zero unresolved findings; screen-reader	FR-020, SC-009	NOT-AUTOMATED (spec tasks T030/T035 unticke

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		gamification set	validation signed		

Spec 015 — Universal content model and learner profile

Delivers page/location anchors for document-sourced citations (without breaking video anchors), four read-only profile API endpoints (`/api/profile`, `/api/profile/activity`, `/api/profile/reports`, `/api/profile/areas/{area}`; routes confirmed in `platform/backend/internal/api/profile.go`) and the `/profile` page: activity feed, per-area reports, a hand-rolled SVG progress chart, empty states and a "retake self-assessment" action.

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
QA-015-01	Anonymous	Open `/profile`; `curl -i <base>/api/profile`	Redirect to login / API 401	FR-013	NOT-AUTOM
QA-015-02	Student	New student (just changed password, no activity), open `/profile`	Four independent explicit empty states (header, activity, reports, chart); no error banner	FR-018, SC-005	TESTED-BY- `profile.spe` empty state TestProfileS
QA-015-03	Student	After some study (lessons, practice answers), open `/profile`	Header shows account age and completion; one reverse-chronological activity feed (newest first) across lessons, answers,	FR-014, SC-003	TESTED-BY- merged tim `profile_tes`

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
			reading positions		
QA-015-04	Student	Scroll the activity feed past the first page	Loads 20 at a time, "load more" follows the server cursor; history size does not slow the page	FR-014, SC-007	TESTED-BY- loads one pa `profile_bou`
QA-015-05	Student	Profile chart section after progress in two or more areas	Inline SVG chart (no external chart library) with area labels inside the SVG; hover/ keyboard reads values	FR-015, FR-017	TESTED-BY- inline SVG c
QA-015-06	Student	Per-area reports on `/ profile`; click "retake self-assessment" on an unlocked area	Link lands on the area test and uses the existing assessment endpoint with the documented body	FR-015, FR-016	TESTED-BY- "the retake l `profile_ret`
QA-015-07	Student	Report row for a LOCKED area	Remaining requirements stated in words; no retake link	FR-016, FR-020	TESTED-BY- area states i
QA-015-08	Student	Retake attempted when the assessment endpoint is unavailable	Explicit failure, not a silent no-op	FR-020	NOT-AUTOM covers the h
QA-015-09	Student			FR-019	

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		Reports endpoint unavailable (operator breaks it) then reopen `/profile`	Only chart and reports sections degrade, the rest still renders		TESTED-BY- unavailable sections"
QA-015-10	Student	In devtools Network look at all `/api/profile*` requests; try `?user=<other>` by hand	Only parameterless reads; another user's data is unreachable; no answer key or held-back area leaks	FR-013, SC-004	TESTED-BY- parameterle TestProfile_ TestProfile_ TestProfile_ TestProfile_
QA-015-11	Student	`/profile`: click an activity row for a reading position, and a document material from an area (`/documents/<slug>/read`)	Row opens the right page/location; a malformed percent escape in the link still renders the feed	FR-001, FR-008 (document links)	TESTED-BY- position hre feed"; docu NOT-AUTOM
QA-015-12	Student	`/profile/medium` ("What has worked for you")	Reachable from navigation; ready state shows suggestion plus evidence and controls; "scoring unavailable" differs from	FR-021 (cross-source usage)	TESTED-BY- FR-021/FR-0 not delivere CLARIFICAT

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
			"no answers yet"; turning suggestions off is one PUT and page then says so		
QA-015-13	Operator	Run `go test ./internal/api/ -run Profile` and `internal/textbooks` anchor tests	Pass; video-only anchor rejection for document content stays; page anchors accepted for document content	FR-001..FR-005, SC-001, SC-002, SC-006	TESTED-BY- `anchor_test`

Spec 016 — Workshop live QA fixes

Delivers fixes for issues found in live QA: every recorded chapter shows a recording entry (and a labelled "no recording" indicator by design), no placeholder "Not in this build" text in transcripts, fewer "Transcriber unsure" flags, relevant-only "Where else this points" lists, no truncated progress labels, completed lessons shown checked on return, practice answers selectable/recorded/survive reload, and chapter 4 rewritten in third person. Several pieces need data work done by the operator (rebuild/restart); see `evidence/n4-016/tasks-016-status.md`.

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
QA-016-01	Student	`/` (curriculum): look at the recordings rail for every chapter	Each chapter with a real video shows a working recording entry that opens; each chapter without a video by	FR-001, FR-002, FR-017, SC-001, SC-010	TESTED-BY- AUTOMATI `frontend/e2 verify.spec` server, need `qa-016-live` E2E_SESSIO `e2e/curricu rails.spec.`

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
			design shows the labelled "no recording" indicator (not an empty slot); indicator is keyboard reachable with an accessible label		
QA-016-02	Student	`/` : expand a second chapter rail with the keyboard (Enter/Space)	Only the first chapter starts expanded; others expand/ collapse by keyboard; each shows its own data	FR-001	TESTED-BY- AUTOMATI `curriculum- rails.spec.` when fewe chapters)
QA-016-03	Student	Open `/ chapters/:slug/ transcript` for EVERY chapter and scroll	No segment shows "Not in this build" or any bracketed build placeholder; unavailable segments show the platform's honest "content unavailable" presentation	FR-003, FR-004, SC-002	TESTED-BY- AUTOMATI Go sweep `p` `backend/int` `placholder` `sweep_test.` (machinery mutations; `n4-016/task` `status.md` `s` SERVED cra been run). page throug chapters
QA-016-04	Student	Transcript page, count segments	At least 50% fewer flags per chapter;	FR-005, SC-003	NOT-AUTO task T040 s

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		tagged "Transcriber unsure" for chapter 01 and compare with the pre-fix figure in `docs/qa/KNOWN-LIMITATIONS.md` L8 (267 of 1,055)	segment count unchanged		T041 untick re-transcrip
QA-016-05	Student	Transcript passage page: open "Where else this points" on several passages across chapters	Only genuinely related passages/ documents/ terms; no bare glossary terms or bookkeeping rows; weak relevance shows an honest empty/low-result state	FR-006, FR-007, SC-004	TESTED-BY- AUTOMATI Go `platform` `pkg/crossre` `floor_test.` `api/crossre` FE empty-st spec; a sam read (T044/ AUTOMATE must be re- the operato policy to sh
QA-016-06	Student	`/progress` at 360, 768, 1280 px widths	First-column labels are complete words/ phrases, never cut mid-word	FR-008, SC-005	TESTED-BY- AUTOMATI `live-verify` (per width;
QA-016-07	Student	`/areas/<slug>`: tick a lesson, reload, return from another page	Ticked lesson shows checked immediately on load; unticked lesson is never shown	FR-009, FR-010, FR-017, SC-006, SC-010	TESTED-BY- AUTOMATI `live-verify` (real server

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
			checked; checkbox is labelled and toggles by keyboard		
QA-016-08	Student	Brand-new student opens an area page	No lesson shown as completed (no false positives)	FR-010	TESTED-BY-AUTOMATI (spec T017c `frontend/sr features/ar` check NOT- beyond `016` `verify.spec`
QA-016-09	Student	`/practice/<slug>`: click an option on a multiple-choice question	Option shows as selected immediately	FR-011, SC-007	TESTED-BY-AUTOMATI `live-verify` (real server
QA-016-10	Student	Submit the answer	Receipt says the answer was recorded with the real result; the "An answer was not recorded" message does not appear under normal conditions	FR-012, SC-008	TESTED-BY-AUTOMATI `live-verify`
QA-016-11	Student	Force a failure (operator pauses the progress service), submit, restore, click "Send again"	Retry actually records (not just repeats the failure)	FR-013	TESTED-BY-AUTOMATI unit/compo `platform/fr app/feature practice-answers.ser` the live pau is NOT-AUT

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
QA-016-12	Student	Answer, navigate away, return, then hard reload	Answer is still shown as recorded	FR-014	TESTED-BY-AUTOMATI `live-verify` ("reload kee
QA-016-13	Professor-admin	Read every section of chapter 4 (`/chapters/<chapter-4-slug>`)	Each section reads as a clear concluded statement in the third person, traceable to the chapter 4 conversation; no first-person "I ..."	FR-015, FR-016, SC-009	TESTED-BY-AUTOMATI mechanical `evidence/n4 green.txt`; t read (T060 NOT-AUTOM drops old sp after an ope rebuild/rest
QA-016-14	Operator	Run `bash scripts/qa-016-live-verify.sh` against an isolated server (not 8087)	All US1, US2, US3, US7 e2e assertions pass	SC-001, SC-005..SC-008, SC-010	TESTED-BY-AUTOMATI `frontend/e2 verify.spec` the operato T014/T020/T unticked in a live run is
QA-016-15	Student	Anonymous practice answers made before sign-in	Not migrated into the account (known gap)	FR-014 spirit	KNOWN-DE `evidence/n4 tasks-016-s` "Not done: a >account m practice ans

Spec 017 — Exhaustive RAG expansion (search, term and section deep links)

Delivers PDF textbook ingestion with licence and location anchors, a reranking stage (default OFF), per-content-type chunking, uniform retrieval across content kinds, and (for users) hits that open real destinations: `/passage/:pid` for `doc_section`/`code`, `/term/:term` for

terms, `/section/:section` for document trees. The Jordan client-research correction (FR-014/015/017) and embedding-model evaluation (FR-016) are operator/research items without a UI.

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
QA-017-01	Student	`/search`: query a phrase that matches a document section, a code passage and a transcript segment	Each hit shows its source label and opens a real page; hits with no destination render as plain text (no dead links)	FR-009, FR-013, SC-005	TESTED-BY-AUTOMAT `platform/backend/pkg/ deeplink_test.go` (serv side), `platform/fronte search-degradation.sp` "a hit links exactly wh server gave it a destin `e2e/live-content.spec` "search finds real pass (real server)
QA-017-02	Student	From a search hit click through to `/passage/<26-char ULID>`	Passage page shows title, locus line, source label, the passage text (code in a code block), "Around this passage" neighbours, and a return link to the search	FR-004 (anchor shown)	TESTED-BY-AUTOMAT partial Karma `platfor frontend/src/app/feat passage/passage.spec` `internal/api/ passages_test.go`; the path in a real browser partly covered by `e2e/ content.spec.ts` "a hit to its passage" (transcr segments)
QA-017-03	Student	`/passage/<unknown id>` and a withheld passage id	Absent state (not a blank page); withheld shows the custody message; unavailable shows retry	FR-011 spirit	TESTED-BY-AUTOMAT partial Karma `passage.spec.ts`; `inter api/ passages_publication_`
QA-017-04	Student	From a search hit for a glossary term click	Term page: title, article, "Where this term appears" occurrences	FR-013	TESTED-BY-AUTOMAT partial Go `platform/ba internal/api/terms_te publication_term_test` Playwright spec exists

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		through to `/term/<name>`	each linking on; return link; absent / withdrawn / unavailable states are distinct		`term/:term` (grep of e2 none) so the page itsel NOT-AUTOMATED in a browser
QA-017-05	Student	`/section/<section>` (use a section name from the Documents tree; verify names via the `GET /api/sections/...` listing)	Left tree "Documents", heading with the section, "On this page" outline; absent section shows an explicit absent state	FR-013	TESTED-BY-AUTOMAT partial Go `platform/ba internal/api/sections_test.go`, `section_roundtrip_tes` Karma `features/secti landmarks.spec.ts`. Br flow NOT-AUTOMATE
QA-017-06	Student	Unknown path e.g. `/passage/` mistyped or `/nonsense`	Not-found page, no silent redirect to home	routing	TESTED-BY-AUTOMAT `e2e/live-content.spec` unrouted deep link re not-found"
QA-017-07	Student	Textbook-sourced hit: check citation	Hit shows book, section/chapter and page (precise anchor); opens the textbook page or passage	FR-004, SC-002	TESTED-BY-AUTOMAT `platform/frontend/e2e textbooks.spec.ts` "ma search shows the citat beside a textbook hit" fixture server); Karma `passage-textbook-citation.spec.ts`
QA-017-08	Student	Search quality sample: 10 known-answer queries for textbook content,	Best answer in the top results for each; no content type crowded out (cross-encoder	FR-008, SC-002, SC-003, SC-005	NOT-AUTOMATED end (benchmark data `plat backend/evidence/retr bench-2026-09-04.tsv`, `platform/gates/prove compose-rerank-wiring` Note `reviews/n2-2026-10-22.md` F4: t

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		transcripts and docs	rerank stays OFF by default)		"11/22 vs 15/22" figure for the cross-encoder is really the in-windo OR benchmark
QA-017-09	Student	Search while one leg is down (answering OFF means no answer leg; semantic/ lexical report separately)	Page shows per-leg chips with different outcomes and an honest degraded banner; never an empty-looking "no results" for an undetermined answer	FR-013	TESTED-BY-AUTOMAT `e2e/search-degradation.spec.ts` (0/1/2 and "degraded" `e2e/inquiry-evidence` "the search legs are re
QA-017-10	Operator	Ingest a new textbook into an already-indexed corpus (`bash scripts/ingest-textbooks.sh` on a COPY of the registry)	Only the new content is indexed; no full rebuild; re-run is idempotent with zero duplicates; licence missing refuses	FR-002, FR-003, FR-011, SC-001, SC-004	TESTED-BY-AUTOMAT `scripts/ingest-textbooks.test.sh`; `pl backend/pkg/chunking` operator run on real c is NOT-AUTOMATED
QA-017-11	Operator	Read each ingestion run's report	Reports ingested sections and an explicit (possibly empty) extraction-failure list; ambiguous chapter structure falls back and is flagged;	FR-005, FR-006, FR-007, SC-007	KNOWN-DEFECT: part `reviews/n2-2026-10-2` (silent loss with `extraction_failures:` (evidence JSON from r names, exit 0 with em Re-verify against the candidate

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
			deterministic across two runs		
QA-017-12	Operator	Spot check the corrected client-research material under `workshop/docs/research/clients/`	No remaining Germany-directory references; claims whose basis was the wrong person are corrected or removed; findings re-checkable	FR-014, FR-015, FR-017, SC-006, SC-008	NOT-AUTOMATED (a h reads the material; do copy content into any file)
QA-017-13	Operator	Embedding-model evaluation report exists and a regeneration of vectors uses the generation-versioned path	Evaluation covers the named candidates; a model change does not break existing generation handling	FR-012, FR-016	NOT-AUTOMATED (ve `docs/chunk-vectors.md` chunk-vector gain is UNMEASURED per `rev n2-2026-10-22.md`)

Notes for the tester

- Spec checkbox state in `specs/*/tasks.md` is stale in both directions (see `reviews/partials/spec013.md` item 6); this checklist was derived from the candidate code and tests, not from tick marks.
- Automated browser specs that use `page.route` mocks prove the UI's behaviour against a described contract, not the real server; they are labelled "partial" above.

Part D

Candidate tree (read-only): `~/Projects/vasic-wt/qa-candidate`. Specs: `specs/<NNN>/spec.md`. Defect sources: `the review reports (by file name)` (names cited per row).

How to read this section

- Roles: anonymous (not signed in), student (`student1`..`student5`), professor-admin (`milosvasic`, `rami`), operator (a person with shell and server-flag access).
- Anonymous visitors get 401 on `/api` and are redirected by `authGuard` to `/login?returnUrl=...`; every route below sits behind that guard (verified in `platform/frontend/src/app/app.routes.ts`: only `/login` is outside it).
- Students start on the initial password and are forced to `/account` first (spec 023). Complete that before running any student row (QA-023-03 to QA-023-06).
- Header nav in the candidate (verified in `app.component.ts`): Curriculum, Chapters, Ask or search, Textbooks, Library, Progress, Profile, Answering, What works for you, Status, and Accounts (administrators only).
- Paths: `FE` = `platform/frontend`, `E2E` = `platform/frontend/e2e`, `API` = `platform/backend/internal/api`. All relative to the candidate tree.
- Many e2e specs need an isolated server or a scratch config (stated in the row). They are never run against port 8087. A skipped e2e is not a pass, so those rows say "partial".
- "verify" in an Expected cell means the spec requires it but the UI wording was not confirmed in code; the tester records what is actually shown.
- Registry reality check (read from the candidate's tracked `curriculum/passages.jsonl`, 34,817 rows): it holds 0 rows for `clients/jordan` and 0 for `education-platform`, and 0 `textbook_section` rows. The QA instance may load a different registry. Rows that depend on this say "verify against the QA instance".
- UX flags (spec 025): all eight default to OFF in `FE/src/app/core/ux-flags.ts`. A tester turns one on per browser with `localStorage.setItem('workshop-ux-flags', JSON.stringify({palette:true}))` and a reload. Ask the operator whether the QA build flipped any default.

Spec 018 — Research library and document re-ingest

Delivers a Library (`/library`, `/library/read`) listing every document under the research tree with formats, a searchable state and a reader, plus a deterministic row-only re-ingest of the five research documents

that had no search rows. Documents in scope: `docs/research/clients/jordan/*` and `docs/research/education-platform/*` (four documents), plus four root research documents.

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
QA-018-01	anonymous	Open `/library` while signed out	Redirected to `/login?returnUrl=%2Flibrary`; `GET /api/library` answers 401	FR-001, FR-004	TESTED- the libra
QA-018-02	student	Sign in, click header "Library"	Library opens; the nav item is marked current on `/library`; reachable in one click from `/`	FR-012, SC-007	TESTED- on /libra
QA-018-03	student	`/library` list	Groups: root research, Jordan client, education platform, education-platform proofs-of-concept. Shows 9 documents and 5 supporting files, counts equal `find` on the tree	FR-001, SC-001	TESTED- `library_` in `E2E/l`
QA-018-04	student	`/library`, inspect each row's format links (Read, HTML, PDF)	Only formats that exist are drawn; a missing format is absent, not a dead link; every drawn link answers 200	FR-001, FR-004, SC-002	TESTED- answers
QA-018-05	student	`/library`, read the searchable badge on each of the 9 documents	Badge reads "Searchable", "Partly searchable" or "Not searchable yet" (copy in `library-copy.ts`). Expect "Partly searchable" plus a "N sections withheld" note on `retrieval-and-grounding` (its `[REDACTED]` headings); the five documents	FR-009, FR-013	TESTED- `library_` label re

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
			with no registry rows ("Jordan" and the four education-platform ones) read "Not searchable yet" until the re-ingest is live. verify against the QA instance		
QA-018-06	student	Click Read on a root research document (for example the ux-research one)	Opens `/library/read` with rendered markdown, heading outline, no raw HTML injected; unsupported constructs are reported, not dropped	FR-005	TESTED- with its
QA-018-07	student	Open the Jordan client document (`clients/jordan`) from the Library	Opens in the reader for any signed-in user (spec default Q2); content renders; no 401	FR-004	NOT-AU look
QA-018-08	student	Open each of the four `education-platform` documents (adaptive assessment, document ingestion, gamification, multi-provider)	Each opens in the reader with headings and tables; HTML and PDF siblings open in a new tab with no 401	FR-004, SC-002	NOT-AU four are
QA-018-09	student	In the reader choose HTML, then PDF	Each opens in a new tab served from `/docs/` using the session; page loads	FR-004, SC-002	TESTED- (isolated
QA-018-10	student	Open `retrieval-and-grounding` in the reader and find the sections	The word `[REDACTED]` is shown exactly as the document has it, nothing reconstructed. Then	FR-009, SC-005	TESTED- platform [REDACTE

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		headed `[REDACTED]`	search for any phrase from near those sections: no passage text for the withheld sections is retrievable		
QA-018-11	student	In the Library open a supporting file: the sample `.epub` under education-platform `poc`	Download link and the words "binary file, not previewed"; no garbled text	FR-004	TESTED- link and
QA-018-12	student	Open a supporting `.py` or `.ts` file from `poc`	Shown as code or plain text, not as HTML	FR-004	NOT-AU
QA-018-13	student	Hand-type `/library/read` with a path-escape id (`..%2F..%2Fetc%2Fpasswd`)	Not-found state; server answered 404; nothing outside the research tree is served	FR-003, SC-008	TESTED- `library.`
QA-018-14	student	On `/library` type a title fragment in the filter box, then a string that matches nothing	List narrows with no network request per keystroke; no match reads "no document matches", visibly different from a load failure	FR-013	TESTED- with no
QA-018-15	student	Load `/library` while the server's docs root is unreadable (operator breaks it on a scratch copy only)	An explicit "could not determine" state naming the cause; never an empty list that looks like "no documents"	FR-002, FR-013	TESTED- `library_` and `Test`
QA-018-16	student	Search (`/search?q=...`) for a term that exists only	At least one hit from `docs/research`; the hit card names the	FR-011, SC-003	KNOWN "MISSIN

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		in a Jordan or education-platform document	source document and links to it in the Library		`library-` registry
QA-018-17	student	On `/library` submit a content query (not a title)	Hits limited to research passages, a count of how many hits were kept, each links to the reader	FR-016	KNOWN library .
QA-018-18	operator	After a live re-ingest, run it a second time (`scripts/ingest-research-docs.sh`, stage then apply) and read `reingest.json`	Second run: 0 new rows, 0 changed rows; changed pre-existing lines = 0; model calls = 0; filename-shape row count reported	FR-006, FR-007, FR-008, SC-004, SC-006	TESTED- `ingest-r` (staging
QA-018-19	operator	Check the stale-sidecar gate: edit a research `.md` on a scratch copy and re-run the wrapper	Wrapper regenerates the sidecar and refuses a sidecar whose source hash does not match	FR-010	KNOWN sidecar_ candida
QA-018-20	operator	`GET /api/library/sources?pids=<csv>` with a known research pid, an unknown pid and a suppressed pid (devtools console or curl with a session)	Known pid resolves to its document; unknown and suppressed look identical; body has no path; more than 50 pids refused	FR-016	TESTED- `library_` `TestLibr`
QA-018-21	student	`/library` and `/library/read` at 360 px width, light and dark	No horizontal page scroll, no axe violations, every control Tab-reachable	FR-004	TESTED- wide: no

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		theme, keyboard only			

Spec 019 — Live corpus promotion, deploy gate, What is live panel

Delivers an additive, reversible web and registry promotion, a deploy gate (G0 to G7) with a verdict file, `GET /api/system/provenance`, and a "What is live" panel on `/status`.

ID	Role	Flow / click path (URL)	Expected result	FR / SC	S
QA-019-01	student	Header "Status" then `/status`	A "What is live" panel with build id and commit, dirty flag, bundle manifest hash, registry digest loaded versus on disk, corpus counts by kind and content type (summing to total), index generation, feature flags	FR-012, SC-004, SC-008	K l r u
QA-019-02	student	Same page, provenance endpoint unavailable (operator stops the endpoint on a scratch run)	Rest of Status still renders; panel says it could not determine, with a cause; never blank	FR-012, FR-013	T f
QA-019-03	anonymous	`GET /api/system/provenance` with no session	401	FR-004	

ID	Role	Flow / click path (URL)	Expected result	FR / SC	S
					N u r
QA-019-04	professor-admin	In a signed-in browser console run `fetch('/api/system/provenance').then(r=>r.json())`	JSON with build, web, registry, corpus, index and features blocks; no URL, hostname, key, token, secret or password anywhere	FR-004, FR-005, SC-006	T p T
QA-019-05	operator	Compare provenance numbers to an independent computation (sha256 of the web `index.html`, registry file hash, row counts)	Equal	FR-004, SC-004	T p T
QA-019-06	operator	Run `scripts/deploy-gate.sh` against the QA stack and read the verdict JSON	Exit 0 PASS only if bundle, commit and registry digests agree on host, container and loaded; each broken check gives exit 1 with a reason; stack down or unreadable gives exit 2, never PASS	FR-007, FR-008, FR-009, FR-013, SC-003	T d
QA-019-07	operator	`scripts/restart.sh --require-verdict <file>` with a stale or forged verdict	Refuses	FR-010, SC-007	K r r
QA-019-08	operator				

ID	Role	Flow / click path (URL)	Expected result	FR / SC	S
		`scripts/promote-web.sh` on a staged real Angular `dist` (dry run, then promote, then promote again, then `--rollback`)	New hashed assets first, `index.html` swapped by one rename, second run reports "already promoted" with no writes, rollback restores the previous index	FR-001, FR-002, FR-003, SC-001, SC-002	T p i v
QA-019-09	operator	`scripts/promote-registry.sh` candidate with all current lines plus new rows: apply, apply again, `--rollback`, `--status --verify`	Pre-existing lines unchanged, backup has a different inode, second apply is a no-op, rollback restores the pre-digest, a tampered journal row is detected	FR-011, SC-005	T r
QA-019-10	operator	`scripts/status.sh`	One informational line `live-vs-built: PASS or FAIL or UNDETERMINED` within 5 s; exit codes unchanged	FR-012 (US5)	K c u
QA-019-11	professor-admin	After the operator promotes a new web bundle, hard-reload any page	Latest UI changes visible; no 404 for assets	SC-001, SC-002	N c

ID	Role	Flow / click path (URL)	Expected result	FR / SC	S
			during promotion		

Spec 020 — Textbook library UI

Delivers `/textbooks` (3 books: OpenAI API Cookbook, LLM Engineer's Handbook, AI Engineering: Building Applications), a book page with chapter outline and sections, a page reader with "PDF page N" citations, search-hit citations, and a Figures view. The tracked catalogue (`curriculum/textbooks/catalogue.json`) lists 179, 513 and 983 sections, 183, 523 and 991 pages, and 0 diagrams for each book.

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
QA-020-01	student	Header "Textbooks" then `/textbooks`	Three cards with title, "operator-directed 2026-10-20" licence basis, section, diagram and page counts. If the QA registry has no textbook rows the card reads "Catalogued, not yet live", never 0. verify against the QA instance (tracked registry has 0 `textbook_section` rows)	FR-006, FR-013, SC-003	TESTED-BY-AUTOM books with live cou `textbooks_test.go:`
QA-020-02	student	`/textbooks` with no catalogue present (operator scratch run)	Determined empty state "no textbooks ingested", distinct from "could not load" with a cause	FR-013	TESTED-BY-AUTOM `textbooks_test.go: TestTextbooksList_`

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
QA-020-03	student	`/textbooks` when the API answers 503 (operator scratch run)	Unavailable state with plain words and a working retry; no raw status code	FR-013	TESTED-BY-AUTOM working retry" (mo
QA-020-04	student	Click a book: `/textbooks/<id>`	Chapter outline with page ranges and section counts summing to the book total; paged section list; a chapter link filters the list; a section opens its page	FR-008, US4	TESTED-BY-AUTOM page text" (synthet `textbooks_test.go:`
QA-020-05	student	Open a page: `/textbooks/<id>/page/<n>`; press next or previous (keyboard too); jump to a gap page	Page text shown, labelled "PDF page N"; next and previous cross gaps left by skipped stub pages; a gap page says "no extractable text on this page"	FR-008, FR-012	TESTED-BY-AUTOM catalogued page wi
QA-020-06	student	Open a page whose text trips the server's filename-shape guard (operator names one if any exists; 32 of 34,817 live rows match the guard)	Plain-words custody message (page cannot be shown here), no error code; citation and navigation still work	FR-010, FR-013	TESTED-BY-AUTOM custody state in pla `title.spec.ts`
QA-020-07	student	On `/textbooks` use	Only textbook hits; each shows		TESTED-BY-AUTOM hit with citation ->

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		the "Search textbooks" box with a term unique to one book	book title, section title and PDF page; click opens that page	FR-007, FR-012, SC-004	
QA-020-08	student	`/search?q=<term from a textbook>`	A textbook hit carries the citation (book, section, PDF page) beside it; the page number equals the sidecar `line_start` for that pid	FR-007, FR-012, SC-004	TESTED-BY-AUTOM beside a textbook h `citation`)
QA-020-09	student	`/passage/<pid>` for a textbook passage, and for a non-textbook passage	Textbook passage shows the citation; the other shows nothing extra	FR-007	TESTED-BY-AUTOM `textbooks_test.go:` page rendering is c `citation.component`
QA-020-10	operator	`GET /api/textbooks/citations?pids=` with a live pid, an unknown pid and a suppressed pid	Live pid gives `{book_id, book_title, section_title, page}`; unknown and suppressed give identical `null` bytes; more than 50 refused	FR-007	TESTED-BY-AUTOM `textbooks_test.go:` `TestTextbookCitati`
QA-020-11	student	Book page then Figures: `/textbooks/<id>/figures` for each of the three books	Catalogue says 0 diagrams per book, so the page must show the determined empty state "No figures were found in this book", not a blank. Figures	FR-009, SC-005	KNOWN-DEFECT: c diagram extraction `figures.component.`

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
			count shown equals the catalogue count		
QA-020-12	student	Figures view with diagrams present (only possible once the operator has run extraction; otherwise use the e2e fixture)	Image, caption or "Figure on PDF page N", page link, quality badge (high or low resolution), honest redraw note; a missing file shows an explicit missing state; no SVG served	FR-009	TESTED-BY-AUTOM `textbooks_diagrams` real figures NOT-AU
QA-020-13	operator	`GET /api/textbooks/<id>/diagrams/<n>/image` headers	PNG only, `nosniff`, restrictive CSP, neutral file name, size cap, 404 for flipped or truncated bytes	FR-009	TESTED-BY-AUTOM `textbooks_diagrams` `TestTextbookDiagra` (serve-time sha opt
QA-020-14	operator	Inspect every `/api/textbooks*` JSON body	No file name, path or `source_ref`; passes the redactor unchanged	FR-010, SC-007	TESTED-BY-AUTOM
QA-020-15	operator	`scripts/ingest-textbooks.sh --dry-run` against a copy of the registry	Pure addition of `textbook_section` rows (expected total 1,675), second run 0 new, missing licence basis stops with exit 2, 0 model calls	FR-001 to FR-004, SC-001, SC-002	TESTED-BY-AUTOM merge is recorded candidate's tracked
QA-020-16	operator			SC-006	NOT-AUTOMATED

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		Retrieval-only benchmark before and after the textbook merge	No previously-hit question now misses		
QA-020-17	student	`/textbooks`, a book page and the page reader at 360 px and with reduced motion, light and dark	No horizontal scroll, no axe violations, no running animation	FR-011	TESTED-BY-AUTOM "reduced motion le

Spec 021 — Learning-medium profile ("What works for you")

Delivers `/profile/medium`, shown from the header "What works for you": per-format performance on the learner's own graded answers with evidence, an advisory recommendation, an override (auto, pinned, off) and a reset. Everything is deterministic and model-free; format-engagement capture is opt-in and OFF by default (`-medium-engagement`).

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
QA-021-01	anonymous	Open `/profile/medium` signed out	Redirect to `/login`; `/api` medium routes 401	FR-013	TESTED-BY- `auth-guard.`
QA-021-02	student	Click header "What works for you"	Page loads under the profile area; nav item marked current	SC-007	TESTED-BY- "reachable f mocks on a
QA-021-03	student	A brand-new student (no	HTTP 200 empty state "nothing recorded yet"	FR-015, US1	TESTED-BY- "no answers `API/medium_`

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		answers) opens the page	with zero counts; not an error and not the "unavailable" state		
QA-021-04	student	After answering a few practice questions (fewer than the per-format minimum) reopen the page	State "insufficient evidence" for each format; no recommendation; no fallback to a popularity default	FR-005, FR-006, US1	TESTED-BY-A `medium_test` real student
QA-021-05	student	With enough graded answers in at least two formats (seeded or long practice) open the page	Ready state: per-format counts, observed versus expected, a suggestion with the evidence list and thresholds used; wording describes performance with a format on your own data, not a learning style label	FR-003, FR-007, FR-017, US1/ US2	TESTED-BY-A "the ready s the controls
QA-021-06	student	Short-answer-only data for a format	Shown as ungraded, not counted as performance evidence	FR-004, US1#3	TESTED-BY-A under `platf`
QA-021-07	student	Page when the scoring store is unreadable (operator scratch run)	Own "unavailable" state with a stated reason; never reads as "no answers yet"	FR-014, US1#5	TESTED-BY-A "scoring una `medium_test`
QA-021-08	student	Controls: choose "I choose a format" and a	Each choice honoured on the very next	FR-009, SC-005	TESTED-BY-A "turning sug

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		format; then "Suggest from my answers"; then turn suggestions off	request; "off" makes the page say so and no derived signal is computed		`medium_test` `TestPrefere`
QA-021-09	student	"Reset my data" with its inline confirm	Preference, consent and event rows for this learner are removed; page returns to empty state	FR-012, SC-008	TESTED-BY- `medium/medi` server-side candidate; N
QA-021-10	student	Open an area page `/areas/<slug>` after a recommendation exists	A one-click hint showing the recommended format and why; no hint and no empty box when there is no recommendation	FR-011, US3	KNOWN-DE section 1 fin route and na not embedd of `FE/src/ap` lists 5 UNWI
QA-021-11	student	Practice deck with the selector set to "flashcard"	Deck shows only flashcard-kind questions	US3#3	KNOWN-DE `filter.comp`
QA-021-12	student	Progress page `/progress`, then `/profile/medium`	Progress shows scores; the medium page is the only place that shows format effectiveness. No link from Progress to the medium page is required by the spec; record whether one exists	FR-011	NOT-AUTOM end to end;
QA-021-13	student	Engagement opt-in: on the medium page	With `-medium-engagement` off the consent control is	FR-016, US4	TESTED-BY- `medium/medi`

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		look for the consent control (default flag OFF)	absent or disabled and engagement posts are refused; verify what the UI shows. With the operator flag on: consent, events accepted only in the closed shape, revoke plus "delete my engagement events"		`engagement.` real server)
QA-021-14	student	Sign in as `student1`, then as `student2`; compare `/profile/medium`	Each sees only their own data; the administrator also gets no learner view of others	FR-002, FR-013, SC-003	TESTED-BY- `api/medium_` `TestAdminGe` `TestHandler`
QA-021-15	operator	No-model guard and determinism	The medium package imports no network or model package; identical inputs give byte-identical output	FR-008, SC-002, SC-006	TESTED-BY- `medium-no-m` `mutations.s` `determinism`
QA-021-16	student	`/profile/medium` axe, keyboard, phone width	No serious or critical violations; controls keyboard operable	FR-019	TESTED-BY- "no automat (mocked)

Spec 022 — On-demand grounded study material

Delivers an explicit-request generator of cited study material (explainer, flashcards, short quiz) that declines rather than fabricates, with budgets, audit and an admin review. It ships OFF (`-materials-enabled=false`). In the candidate there is no route and no nav link for it, so the UI is unreachable (see QA-022-01).

ID	Role	Flow / click path (URL)	Expected result	FR / S
QA-022-01	student	Look for "Study material" in the header; then type `/materials` in the address bar	Per the spec the page should exist and, when generation is disabled, say so and offer no form. In the candidate there is no nav item and `/materials` falls through to the "Not found" page	FR-00 FR-02 US1#
QA-022-02	student	With the operator having mounted the route on a scratch copy: open the materials page while generation is OFF	The page says generation is turned off, offers no request form, no model call happens	FR-02 SC-00
QA-022-03	student	Console: `fetch('/api/materials/provider')`	When disabled or not attached: an honest 503 or `enabled:false` body, never a blank or a fake success	FR-02
QA-022-04	operator	Turn it on: start the server with `-materials-enabled` (`WORKSHOP_MATERIALS_ENABLED=1`), a provider, budgets and the answering gate cleared	Boot refuses bad budgets or a missing provider; with the answering gate not cleared the provider is forced off and materials stay unavailable. No compose entry exists for these flags	FR-02 FR-02
QA-022-05	student	Request material for a covered topic (operator-enabled stub	A job runs; result is "generated" with	

ID	Role	Flow / click path (URL)	Expected result	FR / S
		provider only): form shows provider, locality and a token upper bound; confirm	items, each with a citation chip linking to `/passage/<pid>` and a supporting quote; labelled draft	FR-00 FR-01 US1#
QA-022-06	student	Request a topic the corpus does not cover	Result "declined" with a reason in words; no generated text	FR-00 FR-00 US1#
QA-022-07	student	Request while the provider is down	"Unavailable" state with a code and reset time; no partial text	FR-00 US1#
QA-022-08	student	Request again with the same topic	Served from the learner's own cache, byte-identical, no new model call; a retry with the same idempotency key returns the same job	FR-01 FR-01 SC-00
QA-022-09	student	Exceed the per-request or daily token budget	Refused before any provider call with a clear budget message	FR-01 SC-00
QA-022-10	student	Saved list: open one material and delete it (inline confirm)	The material disappears; another student never sees it (404, not 403)	FR-00 FR-02
QA-022-11	professor-admin	Material review page (`/materials/admin` in the unmounted routes)	List with provenance, verification results, decline reasons, token usage; students refused	FR-02 US6
QA-022-12	operator	Remote provider configured: request without acknowledgement, then with	Without ack: `400 remote_ack_required`, 0 provider calls; each remote attempt writes an audit line	FR-01 FR-01 SC-00
QA-022-13	operator			FR-00

ID	Role	Flow / click path (URL)	Expected result	FR / S
		Anonymous request to `POST /api/materials` carrying an `X-Session` header	401, never served under a client-chosen owner key	
QA-022-14	operator	Fabrication matrix and suppressed canary	Every invented pid, non-verbatim quote, out-of-set citation gives declined; a suppressed passage never reaches a prompt or the output	FR-00 FR-01 SC-00 SC-00

Spec 023 — Roles and accounts (professor administrator, student)

Delivers an enforced split: `/api/admin/*` is administrator-only, students are forced to change the initial password at `/account`, a visible role badge, `/account` and `/admin/users`. Accounts: administrators `milosvasic` and `rami`; students `student1` to `student5`. Do not write any password into this checklist.

ID	Role	Flow / click path (URL)	Expected result	FR /
QA-023-01	anonymous	Open `/`, `/admin/users`, `/account` signed out; console `fetch('/api/account')`	Each page redirects to `/login?returnUrl=...`; the API answers 401	US1 SC-0
QA-023-02	anonymous	`/login` with a wrong password; then repeat 6 times within a minute (use a throwaway name, not a real account)	Plain failure message each time; the sixth attempt in a minute is rate-limited (5 per minute per source address)	Risk sect
QA-023-03	student	First sign-in as a student on the initial password (credential from the operator)	Lands on `/account`; banner "you must change your password" (`account-must-change`); role badge reads "Student"; no "Accounts" nav link and no administrator section	US3 US4

ID	Role	Flow / click path (URL)	Expected result	FR /
QA-023-04	student	Still on the initial password, visit `/progress`, `/chapters`, `/admin/users`; in the console `fetch('/api/areas')`	Every page returns to `/account`; the API answers 403 with error `password_change_required`	US3 FR-0
QA-023-05	student	On `/account` fill current, new and confirm password: mismatch; too short; contains the user name; then a valid one	Mismatch and weak values show an error that names the required minimum and never echoes the password; a valid change sends the student to `/login?reason=password_changed` with a notice "password was changed"	US3 FR-0 FR-0
QA-023-06	student	Sign in with the new password	Lands on `/` (learner surface); badge "Student" with `data-role=user`; no administrator link anywhere (`/`, `/progress`, `/account`)	US3 US4
QA-023-07	student	Type `/admin/users` in the address bar	Redirected to `/account?denied=admin` with a visible refusal message (`account-denied`); no list shown	US4 D5
QA-023-08	student	Console as a settled student: `fetch('/api/admin/users')`, `fetch('/api/admin/does-not-exist')`, `fetch('/api/account')`	First two answer 403 `forbidden` (a nonexistent admin path is not an existence oracle); `/api/account` 200 with `role_label student` and no hash	US1 FR-0 SC-0
QA-023-09	student	Same console probes using dot-segment and case tricks (`/api/admin/./users`, `/API/ADMIN/users`, `/api/admin%2fusers`)	All refused (403 or 404), never 200	US1 SC-0
QA-023-10	professor-admin	Sign in as `milosvasic` (and separately `rami`)	Badge "Administrator" (`data-role=admin`); header shows "Accounts"	US4

ID	Role	Flow / click path (URL)	Expected result	FR /
QA-023-11	professor-admin	Header "Accounts" then `/admin/users`	Lists the seven accounts (`milosvasic`, `rami`, `student1` to `student5`) with a role chip each; `rami` and `milosvasic` admin, students user; no password hashes in the page or the network response	US4 FR-0
QA-023-12	professor-admin	Click the username then `/account`	Role, what-you-can-do text, change-password form and the administrator section (`account-admin-section`); Logout returns to `/login`	US4
QA-023-13	professor-admin	`/admin/users` when the accounts API fails (operator scratch run)	Error state with a retry (`admin-users-retry`), not an empty table	US4
QA-023-14	operator	Run `scripts/provision-accounts.sh` twice on a throwaway store; then with the secrets file at mode 0644 and with the secret missing	Second run changes nothing; a student's changed password is never reset; a missing secret or non-0600 file exits 2 and prints no secret; a weak initial password fails closed naming the minimum	US2 SC-0 FR-0 FR-0
QA-023-15	operator	Role change of `rami` from user to admin during provisioning; then check the old `rami` browser session	Role becomes administrator; previously issued sessions are revoked and give 401 on admin routes	D2
QA-023-16	operator	Start with `WORKSHOP_PASSWORD_MIN_LEN` set to `abc`, `5` and `73`	Each refuses to start or provision with a message naming the allowed range 6 to 72; the default 8 is not silently substituted	FR-0
QA-023-17	operator	Confirm the administrator account is spelled `rami` (the operator once wrote	The account list matches the roster in D2; no	D2, FR-0

ID	Role	Flow / click path (URL)	Expected result	FR /
		"rtami") and that the initial student password was supplied by env or a 0600 file only	password in any file, log or evidence	
QA-023-18	student	As a settled student open `/status` and `/api/index/status`	Spec leaves this open (Q5): record what a student sees; no decision exists	Q5
QA-023-19	student	`/account`, `/admin/users` (as admin) with keyboard and axe	No serious or critical violations	US4

Spec 024 — Answering enablement and provider settings UI

Delivers `/settings/providers` (nav "Answering"), a read-only page showing the answering state, running provider, thresholds, gate verdict, verdict history, residency note, provider catalogue and (admin only) audit summary; plus the "answering is off, and why" banner on `/ask` and a two-sided measured gate for turning answering on. QA instance: answering OFF (`-answer-provider none`, the compose default), hosted providers OFF.

ID	Role	Flow / click path (URL)	Expected result	FR / S
QA-024-01	anonymous	Open `/settings/providers` signed out	Redirect to `/login?returnUrl=...`	FR-00
QA-024-02	student	Header "Answering" then `/settings/providers`	Title "Answering & providers"; lede says the page only shows state and cannot change it; the only button is Refresh; state card title "Answering is off" with the reason sentence; "Search and browsing are unaffected in every	FR-00 SC-00 US1#1

ID	Role	Flow / click path (URL)	Expected result	FR / S
			state."; link "Go to Ask"	
QA-024-03	student	Same page, "Running provider" card	Provider none, model "none", endpoint "none", locality badge with text (local, LAN, remote or unverified), "Remote opt-in: off"	FR-00 US1#2
QA-024-04	student	"Thresholds" card	"Calibrated yes or no" plus a plain "this means" sentence, minimum score and margin, support and question check kinds	FR-01
QA-024-05	student	"Gate verdict" card and "Verdict history"	With no verdict: "No measured verdict exists. Answering can only be turned on by a passing, measured verdict for the exact configuration."; history empty state (not a blank)	FR-00 FR-01 US4#1
QA-024-06	student	"Residency note" card	Shows the operator-authored note, or an explicit absent state when none; an agent never drafts it (T078)	FR-01
QA-024-07	student	"Provider catalogue" table	Every registered provider exactly once with kind and upstream; permanent stubs marked not selectable; unset credential reads "credential: not present" showing the variable NAME only, never a value, length or prefix	FR-00 FR-00 US1#3

ID	Role	Flow / click path (URL)	Expected result	FR / S
QA-024-08	student	"Audit" card as a student	Administrators-only sentence; no audit rows in the response	FR-00 US1#5
QA-024-09	professor-admin	Same page as an administrator	Audit card shows per-provider attempt counts and recent attempt metadata (never prompt, passage or answer text), or "audit log not available on this build"	FR-00 US1#5
QA-024-10	student	The six states: observe the one the QA instance shows (expected `off_by_config`, "Answering is off"). The other five are operator-forced on a scratch server: `off_gate_not_cleared` ("Answering was forced off"), `suspended_ingest` ("paused for ingest"), `degraded_provider_unhealthy` ("on, but its provider is not responding"), `on_cleared` ("Answering is on"), `measuring` ("MEASUREMENT MODE: answers are not served to learners")	Each state has exactly one title and one sentence, with an icon plus text, not colour alone	FR-00
QA-024-11	student	Make the status unreadable (operator stops the endpoint on a scratch run)	"The answering status could not be read" with a Try again button; nothing says answering is on or off	FR-01
QA-024-12	student	Refresh button after the operator changes state on a scratch server	Page reloads; a polite announcement only when the state changed, silent on first load	FR-01

ID	Role	Flow / click path (URL)	Expected result	FR / S
QA-024-13	student	`/ask` (or `/search`, Ask mode) and submit a question while answering is OFF	The unavailable state with the banner "answering is off, and why", a link "See why on the Answering & providers page", and "Searching and browsing are not affected"; the search side keeps working	FR-01 US1#1 FR-01
QA-024-14	student	Console `fetch('/api/ask?q=test')` and then run a normal search	Ask answers 503 unavailable with reason `no_provider`; search results unchanged	FR-01 SC-00
QA-024-15	operator	Why Ask is OFF in QA: check `-answer-provider` in the running command line and `GET /api/answering/overview`	State `off_by_config`, provider `none`, compose default stays `none`	FR-01 SC-00
QA-024-16	operator	On a scratch server start with `-answer-provider ollama` and no PASS verdict	Boots with answering forced off, state `off_gate_not_cleared`, a structured log line with the decision	FR-00 FR-02 SC-00
QA-024-17	operator	`scripts/answering-enable.sh` with no or stale verdict, then with a matching PASS verdict; then `scripts/answering-disable.sh` and `scripts/answering-postcheck.sh`	No/stale: non-zero exit; matching PASS: prints the single command to apply; disable returns to `none`; postcheck confirms search unchanged	FR-01 US5, SC-00
QA-024-18	operator	Fabrication regression fixtures F1, F2, F3 and the evidence-demand layer	Not user-visible while answering is OFF; F1/F2/F3 produce no `answered` outcome in the harness	FR-01 SC-00
QA-024-19	operator	Hosted provider rows: confirm the page says remote opt-in is required,	Hosted rows show remote locality and "remote opt-in off"; per-request	FR-00 US4#2

ID	Role	Flow / click path (URL)	Expected result	FR / S
		and that no hosted provider is enabled	acknowledgement prompt exists for when enabled	
QA-024-20	student	`/settings/providers` at phone width, axe, keyboard	No horizontal page scroll (table scrolls inside its own region), no serious or critical violations	FR-01

Spec 025 — UX research wave (palette, peek, view transitions, ask-stream stages, copy-link, accessibility, motion)

Delivers eight features, each behind a latent runtime flag that defaults OFF: `askStream`, `practiceConfidence`, `comfort`, `mediaSession`, `momentLinks`, `palette`, `peek`, `railRelevance`. Turn a flag on in the browser (see top of file) before each group; each group also needs a flag-OFF check.

ID	Role	Flow / click path (URL)	Expected result	FR / SC	S
QA-025-01	operator	Read which flags the QA build enables: in the browser console `localStorage.getItem('workshop-ux-flags')`; ask which build defaults were flipped in `core/ux-flags.ts`	All eight default OFF; there is no env or compose way to enable them (backlog); a flag is per browser	FR-020	T F f a
QA-025-02	student	With every flag OFF browse `/`, `/search`, a chapter transcript, a passage	No palette button, no Copy link control, no stage list, no peek popup, no rail relevance layer; layout identical to before	FR-020	T p l c s

ID	Role	Flow / click path (URL)	Expected result	FR / SC	S
QA-025-03	student	`askStream` ON, answering OFF: `/ask`, submit	Existing honest unavailable state and banner; no stage list, no perpetual spinner	FR-001, US1	T E " u h a
QA-025-04	operator	`askStream` ON with answering ON on a scratch stub server (cannot be done on the QA instance): submit a question	Stage list shows exactly the phases the server sent (queued, retrieving, generating, verifying) with server elapsed times; collapses to a summary after the result; never a phantom stage	FR-001, FR-002, SC-001	T p s s o p
QA-025-05	operator	Same setup: a question that declines; a withheld event; one dropped connection; a second drop	Decline renders with its reason; withheld discards any draft; one reconnect then replay without duplicate stages; second drop gives an explicit retry state, never a spinner	FR-004, FR-006	T p s d s p
QA-025-06	operator	Same setup with a screen reader or live-region inspector	Exactly one polite `aria-live` region announces	FR-003, SC-003	T p s a

ID	Role	Flow / click path (URL)	Expected result	FR / SC	S
			stage changes; the answer body is not a live region; no announcement per heartbeat		n a
QA-025-07	operator	Optional draft preview ON (default OFF), stub server	Raw text under a permanent "unverified draft" label, replaced or removed when the verified result arrives (removed with a message on refusal)	FR-005	T p f p (
QA-025-08	student	`practiceConfidence` ON: `/practice/<area>` answer a question	Per spec: reveal must wait for a confidence choice; self-explanation optional; high-confidence-wrong outcome in neutral wording	FR-007, FR-008, US2	K c s it (T c c f c
QA-025-09	student	`comfort` ON and OS "reduce motion" ON: navigate between top-level pages, open a modal	No slide or transform; at most a short opacity fade (150 ms or less); final states unchanged	FR-009, FR-010, SC-004	T p m " o r (C
QA-025-10	student	`comfort` OFF and OS "reduce motion" ON (the default QA state)	Legacy blanket rule still governs; page	FR-010	K c s

ID	Role	Flow / click path (URL)	Expected result	FR / SC	S
			works; verify nothing animates		it r r b E m C l
QA-025-11	student	`comfort` ON with OS "increase contrast" and with "reduce transparency"	High-contrast token set (muted text and borders take the text colour); scrim opaque, no backdrop blur; axe contrast clean on the four main pages	FR-009, US3#3	T p m a t i e
QA-025-12	student	`comfort` ON, normal motion: click between Curriculum, Chapters, Progress; then open a transcript	Short view transition (200 ms or less) between top-level routes; none on the transcript load; browsers without the API behave as before	FR-012, US3#4/5	T E m r t r a i c p
QA-025-13	student	Open a long transcript (1,000 or more rows), scroll fast, follow a `?t=` link and a `#p-<pid>` link	Smooth scrolling; deep links still land on the right row	FR-011, SC-005	T p t (a f d
QA-025-14	student	`mediaSession` ON: play a recording `/chapters/<slug>`; open the OS or browser media	Overlay shows the recording title and	FR-013, US4	T p s

ID	Role	Flow / click path (URL)	Expected result	FR / SC	S
		overlay (a real device, not emulated)	current chapter; media keys seek and jump chapters; leaving the page clears it		n t A
QA-025-15	student	`momentLinks` ON: `/chapters/<slug>/transcript`, press "Copy link to this moment"; open the copied URL in a private window	Polite confirmation "Link copied."; the URL lands on the same passage or time with the arrival announcement	FR-014, SC-006	T E " o la p
QA-025-16	student	Existing `?t=<seconds>` and `#p-<pid>` links, with the flag on and off	Unchanged behaviour	FR-014	T E " f u
QA-025-17	student	`/passage/<pid>` for a normal and a withheld passage; and a link with an unknown pid	Normal: copy-link control; withheld: no copy control; unknown pid: an explicit vanished-pid explanation, not the top of the page	FR-014	T p l p p p c s h
QA-025-18	student	Clipboard blocked (browser permission denied or insecure origin) then Copy link	The URL is shown selected in a read-only field	FR-014	T E " U r
QA-025-19	student	`palette` ON: press Ctrl+K (Cmd+K on Mac) on any page; type "progress"; arrow down; Enter	Palette opens with input focused (ARIA combobox), results	FR-015, SC-007, US6#1	T E " i "

ID	Role	Flow / click path (URL)	Expected result	FR / SC	S
			grouped with text labels, Enter navigates, Escape closes and returns focus to the previously focused element		E r
QA-025-20	student	Palette: close it then press Tab repeatedly	Focus moves on into the page; no focus trap	FR-015	T E f
QA-025-21	student	Palette: use the visible button in the header (mouse or touch), no shortcut	Opens the same palette	US6#5	T E v /
QA-025-22	student	Palette: untick the shortcut toggle inside it; reload; press Ctrl+K; then use the button	Shortcut does nothing, the choice survives reload (WCAG 2.1.4), the button still opens it	FR-015, US6#5	T E " c
QA-025-23	student	Palette: choose "Ask or search for ..."	Hands the typed words to `/search`	US6#4	T E o t
QA-025-24	student	`peek` ON: on `/search?q=...` Tab to a hit; hover another; hover a citation chip in a transcript cross-reference	Preview with passage text and source opens on focus or hover; Escape closes without moving focus; the pointer can move onto	FR-016, US7#1/2	T E " p W " p p

ID	Role	Flow / click path (URL)	Expected result	FR / SC	S
			the popup and it stays; no `title` tooltip		
QA-025-25	student	Peek on an answer citation whose passage is withheld	Popup shows the withheld state, never text	FR-016, US7#3	T E v s a
QA-025-26	student	Peek on a touch device or touch emulation	No popup on hover; tap navigates to `/passage/<pid>`	US7#5	T E t o (
QA-025-27	student	`railRelevance` ON: open `/chapters/<slug>/transcript?q=<term>` then remove `?q=`	With a query the rail shows where it is answered, replacing (not stacking on) the uncertainty layer, with summary text and legend changing together; clearing restores the original layer	FR-017, US8	T E r q R l q l
QA-025-28	student	Rail with a query that has no hits in this recording	Explicit "no matches in this recording" note, not a blank rail	US8#5	T E r i e
QA-025-29	student	Keyboard only through the app shell: first Tab stop, skip link, nav, search combobox, a chapter card, transcript "jump to timestamp", a citation	Skip link first and moves focus into main; every control reachable and	FR-019	T E r li c ju

ID	Role	Flow / click path (URL)	Expected result	FR / SC	S
			operable by Enter or Space; focus ring always visible		c t k
QA-025-30	student	Axe scan equivalents: main pages in light and dark theme	No axe violations	FR-019	T E r p d
QA-025-31	student	Mobile widths: 360, 390 and tablet width on `/`, `/search`, a transcript, `/settings/providers`, `/library`, `/textbooks`	No horizontal page scroll; controls reachable; palette and Copy link usable by touch	FR-019	T p r h a s h p a o
QA-025-32	student	OS "reduce motion" ON for the whole session: visit the main pages, open a modal, toggle the rail, play the achievement toast	Nothing slides or bounces; information unchanged	FR-009, SC-004	T p m d t r (
QA-025-33	operator	No new third-party runtime dependency	`package.json` gains none (Angular CDK not added)	FR-018, SC-008	N a it n f
QA-025-34	operator	Unplanned backend change on the ux branch	`pkg/answer/http.go` `vetResult` change is described in the backlog; confirm it is	FR-004	K c s (u c h

ID	Role	Flow / click path (URL)	Expected result	FR / SC	S
			intended and reviewed		

Spec 026 — Spec reconciliation (specs 011 to 017)

A documentation-only feature: contradictions between specs 011 to 017 are resolved by edits to the spec files. Nothing is visible in the SPA; the operator checks documents and a few commands.

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
QA-026-01	operator	In `specs`, count the marker `Reconciled by 026 on 2026-10-20` in each of 011, 012, 013, 014, 015, 016, 017 (`grep -rc`)	At least 1 in each of the seven directories (measured by the reviewer: 7, 27, 20, 14, 15, 55, 6)	SC-001	NOT-AUTO gate asserts runs the gr are in `umbre guard-027-` "Stale expo
QA-026-02	operator	`git status` after the 026 work (or `git log` of the spec commits)	Only files under `specs/` changed; nothing under `workshop/` or product code	SC-003	NOT-AUTO human mu
QA-026-03	operator	Open the Open Decisions table in `specs/026-spec-reconciliation/spec.md` and compare O-2 to O-8 and the crossref floor with `docs/OPERATOR-DECISIONS-2026-10-21.md` D-08	Each is recorded as ratified 2026-10-21; the 013 spec "Clarifications Needed" and T049 no longer show O-2 open	FR-014	KNOWN-DE `completion- specs-018-0` section 4 ("S O-2..O-7 rat 2026-10-21 tasks still sh O-8 recorde spec")
QA-026-04	operator	Check that task `T012` of 026 names `wire_audit.go`,	Those files exist, or the	FR-013	KNOWN-DE `completion- specs-018-0`

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
		`wire_lan.go`, `wire_rerank.go`	task text is corrected		section 4 ("C wire_audit. wire_lan.go wire_reran exist"); con `umbrella-guard-027-2` section B "C corrections
QA-026-05	operator	Feed the reconciled 012 sidecar example to the real ingest decoder; check `(textbook_section, textbook)` is an accepted pair	Decodes; every `sections[]` key is a declared field	FR-001 to FR-004, SC-002	TESTED-BY- AUTOMATI `platform/ba` `ingest-tran` and `scripts` `textbooks.t` exercise the decoder wi rows; the fi comparison a documen
QA-026-06	operator	Check practice-bank baselines: 292 questions in 40 files, 113 medium, 179 hard, 0 easy (re-run the measurement command in `specs/026-.../research.md` section 5)	Same figures at the candidate's commit, or the spec is updated	FR-008	NOT-AUTO command l spec's resea human mu
QA-026-07	operator	Check the achievement event contract exists: `specs/013-adaptive-assessment-scoring/contracts/achievement-events.md` and the 013 tasks that emit the three kinds	File exists; each kind has trigger, `coalesceKey` and payload; replay endpoint `GET /api/progress...` present	FR-005 to FR-007	NOT-AUTO document c runtime sic by the 013 s outside this

Spec 027 — Research gap closure

A specification only (all 50 tasks unticked): a register `contracts/requirement-register.md` (RG-01..RG-48), a 33-row tooling decision table and 22 NEEDS-OPERATOR entries. No SPA surface. The scripts it plans (`verify-research-register.sh`, `report-tooling-drift.sh`, `test-verify-research-register.sh`) do not exist in the candidate or the umbrella `scripts/` directory.

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
QA-027-01	operator	Open `specs/027-research-gap-closure/contracts/requirement-register.md`	48 rows RG-01..RG-48, no gaps or duplicates; every row has one disposition from the closed vocabulary	US1, FR-001	NOT-A `script resear planne review counts `umbrel guard- sectio
QA-027-02	operator	Run the register check, delete a Decision cell in a throwaway copy, and corrupt the file	Exit 1 naming the row; exit 2 for the corrupt file; never 0	US1	NOT-A script the ca `script resear
QA-027-03	operator	Compare the register to the research documents' ranked tables (#1 to #23) and section 9 gaps	Zero research rows missing from the tooling table	US1#3	NOT-A human checks `guard- verifie end
QA-027-04	operator	Read the 027 claims that 013 still lacks the repeat-attempt detector, integrity reading and 2PL shadow fit (RG-13, RG-14, FR-006 to FR-010)	These now exist in 013 (commit `832b020`, tasks T050 to T053); 027 must be corrected so it builds only the policy	FR-006 to FR-010	KNOW `umbrel guard- B1 ("Sp Phase derive RG-13/ true")

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
			and operator report		
QA-027-05	operator	Check the register line citations (REQ-2 at `:426`, refusal at `pdf_textbook_sections.py:225`) and RG-02's "prove script per tsv" wording	Citations point at the right lines; RG-02 notes only 001, 002 and 006 have a prove script	RG-02, RG-21	KNOW `umbrel` `guard-` B2 and RG-02
QA-027-06	operator	Answer the NEEDS-OPERATOR table (NO-1 to NO-13 and NO-J1..J5) in an `OPERATOR-DECISIONS-<date>.md` file	Each id has an answer or a recorded default; no task tagged with an NO- id starts before it is answered (for example NO-1 whether a local OCR pass counts as a vision pass; NO-5 repeat-attempt policy; NO-9 residency copy and legal review)	US8, FR-036	NOT-A operat nothin answe
QA-027-07	operator	Run the planned `scripts/textbook-draft-outline.sh` twice on the textbook catalogue	Byte-identical output, zero model or network calls, nothing written to the registry	US4, FR-016	TESTE AUTOM the scr candid `textbo` `outlin` `n2-202` lists a

ID	Role	Flow / click path (URL)	Expected result	FR / SC	Status
			or `curriculum/learning/`		its test located
QA-027-08	operator	Scanned PDF ingest: try a scanned (image-only) PDF with and without `--ocr`	Refused with the "no text layer" reason unless the operator opted in; with opt-in, sections marked OCR-derived	US3, FR-011	KNOW `n2-202` `ocr` ca scripts textbo token
QA-027-09	operator	Quality gates the spec plans (quiz-bank validity, WER and CER, retrieval MRR and nDCG, token usage)	Each exits 0, 1 or 2 with a paired mutation proof	US6	NOT-A such g found or `scri` quiz, w token-

9. Cross-cutting checks

Run these on at least the home page, an area page, a lesson, the practice page, the textbook reader and the Library.

ID	Check	How	Expected	Status
QA-X-01	Keyboard only	Tab through each page from the address bar; Enter / Space activate; Esc closes dialogs	Every control reachable and operable in a visible order; a visible focus ring; no keyboard trap; the toast / modal returns focus	NOT-AUTOMATED (partial: Playwright a11y specs under `platform/frontend/e2e` cover some pages)

ID	Check	How	Expected	Status
QA-X-02	Screen-reader names	Use a screen reader or the browser accessibility tree on the nav, practice card, toast region	Landmarks and live regions announce changes once; buttons have names	NOT-AUTOMATED
QA-X-03	Reduced motion	Set OS "reduce motion" (or devtools: emulate `prefers-reduced-motion: reduce`), earn a badge / level up	No decorative motion; the modal and toast still appear and are readable	NOT-AUTOMATED (Karma specs cover the motion switch)
QA-X-04	Light / dark and contrast	Toggle the theme; read body text, badges, disabled states	Text and controls remain legible in both themes	TESTED-BY-AUTOMATION: `platform/gates/verify-served-contrast.sh` (partial: needs a browser run)
QA-X-05	Mobile widths	Resize to 360, 390, 768 px (or a phone)	No horizontal page scroll; nav usable; practice card, reader and tables readable	NOT-AUTOMATED
QA-X-06	Zoom 200 % and large text	Browser zoom 200 %	Layout reflows; nothing clipped	NOT-AUTOMATED
QA-X-07	Anonymous access	Open the URL in a private	Redirected to sign-in / `401` on the API;	TESTED-BY-AUTOMATION: `platform/`

ID	Check	How	Expected	Status
		window; try `/api/areas`, a lesson URL	no content leaks	`backend/internal/api` auth tests (this build: anonymous `/api/areas` answered 401)
QA-X-08	Deep-link reload	Open a deep link (`/area/<id>`, `/term/<t>`, `/chapters/<c>`) in a new tab after sign-in	The page renders directly	NOT-AUTOMATED
QA-X-09	Third-party recording privacy	Browse chapters	No full real name of the recorded third party appears in served pages	NOT-AUTOMATED (gate `verify-source-ref-exposure.sh` passed on the candidate tree)

10. Triage sheet

Copy this table into a spreadsheet or a note; one row per finding. Severity: **S1** blocks the flow or loses data / leaks; **S2** wrong result with a workaround; **S3** cosmetic or wording. Always record the build (the `source_commit` in `GET /api/health`) so a finding can be matched to a candidate rebuild.

ID	Date	Tester	Build (source_commit)	Role / account	Flow (checklist ID)	Expected	Actual
F-001							
F-002							
F-003							